Chapter 6. Inner Product Spaces

Lengths, angles, adjoints and the spectral theorem

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6.1 Inner products and norms

Motivation. Lengths and angles need more than vector-space structure. An inner product, modelled on the dot product, supplies them over $\mathbb{R}$ and $\mathbb{C}$, together with the Cauchy–Schwarz and triangle inequalities, and with the notion of orthogonality that drives the rest of the chapter.

Throughout, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$, and $\bar{c}$ is the complex conjugate.

Definition (Inner product)

An **inner product** on V assigns to $x, y \in V$ a scalar $\langle x, y\rangle \in \mathbb{F}$ such that for all $x, y, z \in V$, $c \in \mathbb{F}$:
(a) $\langle x + z, y\rangle = \langle x, y\rangle + \langle z, y\rangle$;
(b) $\langle cx, y\rangle = c\langle x, y\rangle$;
(c) $\overline{\langle x, y\rangle} = \langle y, x\rangle$;
(d) $\langle x, x\rangle > 0$ if $x \neq 0$. V with an inner product is an **inner product space**: **complex** if $\mathbb{F} = \mathbb{C}$, **real** if $\mathbb{F} = \mathbb{R}$.

By (a), (b), $\langle \sum a_i x_i, y\rangle = \sum a_i \langle x_i, y\rangle$. Over $\mathbb{R}$, (c) reads $\langle x, y\rangle = \langle y, x\rangle$.

Example 1

The **standard inner product** on $\mathbb{F}^n$: $\langle x, y\rangle = \sum_i a_i \bar{b}_i$ for $x = (a_i)$, $y = (b_i)$; for $\mathbb{F} = \mathbb{R}$ it is the dot product $x \cdot y$.

Example 2

If $\langle \cdot, \cdot \rangle$ is an inner product and $r > 0$, then $r\langle x, y\rangle$ is one; for $r \leq 0$, (d) fails.

Example 3

On $C([0,1])$ (real), $\langle f, g\rangle = \int_0^1 f(t)g(t)\,dt$ is an inner product; (d) holds because $f^2 \geq 0$ is continuous and not identically 0 when $f \neq 0$.

Definition (Conjugate transpose)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $A^* \in \mathrm{M}_{n\times m}(\mathbb{F})$ has $(A^*)_{ij} = \overline{A_{ji}}$ (the **adjoint**); for real A, $A^* = A^t$.

Example 4

$\begin{pmatrix} i & 1+2i \\ 2 & 3+4i \end{pmatrix}^* = \begin{pmatrix} -i & 2 \\ 1-2i & 3-4i \end{pmatrix}$.

Example 5

On $\mathrm{M}_{n\times n}(\mathbb{F})$, $\langle A, B\rangle = \operatorname{tr}(B^*A)$ is an inner product (the **Frobenius inner product**): $\langle A, A\rangle = \sum_{i,k} |A_{ki}|^2 > 0$ for $A \neq O$.

The space H of continuous complex functions on $[0, 2\pi]$ with $\langle f, g\rangle = \frac{1}{2\pi}\int_0^{2\pi} f(t)\overline{g(t)}\,dt$ is an inner product space.

Theorem 6.1

For $x, y, z \in V$, $c \in \mathbb{F}$:
(a) $\langle x, y+z\rangle = \langle x, y\rangle + \langle x, z\rangle$;
(b) $\langle x, cy\rangle = \bar{c}\langle x, y\rangle$;
(c) $\langle x, 0\rangle = \langle 0, x\rangle = 0$;
(d) $\langle x, x\rangle = 0$ iff $x = 0$;
(e) if $\langle x, y\rangle = \langle x, z\rangle$ for all x, then $y = z$.

Proof.

(a) $\langle x, y+z\rangle = \overline{\langle y+z, x\rangle} = \overline{\langle y, x\rangle} + \overline{\langle z, x\rangle} = \langle x, y\rangle + \langle x, z\rangle$.

(b) $\langle x, cy\rangle = \overline{c\langle y, x\rangle} = \bar{c}\langle x, y\rangle$.

(c) $\langle 0, x\rangle = \langle 0 \cdot 0, x\rangle = 0$, then conjugate.

(d) From (c) and axiom (d).

(e) $\langle x, y-z\rangle = 0$ for all x; take $x = y - z$ and use (d). ■

An inner product is linear in the first and **conjugate linear** in the second variable.

Definition (Norm)

$\|x\| = \sqrt{\langle x, x\rangle}$ is the **norm** (length) of x.

Example 6

On $\mathbb{F}^n$, $\|(a_1, \dots, a_n)\| = \left(\sum |a_i|^2\right)^{1/2}$, the Euclidean length.

Theorem 6.2

For $x, y \in V$, $c \in \mathbb{F}$:
(a) $\|cx\| = |c|\|x\|$;
(b) $\|x\| = 0$ iff $x = 0$; always $\|x\| \geq 0$;
(c) (Cauchy–Schwarz) $|\langle x, y\rangle| \leq \|x\|\|y\|$;
(d) (triangle inequality) $\|x + y\| \leq \|x\| + \|y\|$.

Proof.

(a) $\|cx\|^2 = c\bar{c}\langle x, x\rangle$.

(b) Theorem 6.1(d).

(c) Trivial if $y = 0$. Otherwise, for $c = \langle x, y\rangle / \langle y, y\rangle$,

$$0 \leq \|x - cy\|^2 = \langle x, x\rangle - \bar{c}\langle x, y\rangle - c\langle y, x\rangle + c\bar{c}\langle y, y\rangle = \|x\|^2 - \frac{|\langle x, y\rangle|^2}{\|y\|^2}.$$

(d) $\|x+y\|^2 = \|x\|^2 + 2\,\mathrm{Re}\langle x, y\rangle + \|y\|^2 \leq \|x\|^2 + 2|\langle x, y\rangle| + \|y\|^2 \leq (\|x\| + \|y\|)^2$ by (c). ■

Example 7

On $\mathbb{F}^n$ with the standard inner product, (c) and (d) read

$$\left|\sum a_i \bar{b}_i\right| \leq \left(\sum |a_i|^2\right)^{1/2}\left(\sum |b_i|^2\right)^{1/2}, \qquad \left(\sum |a_i + b_i|^2\right)^{1/2} \leq \left(\sum |a_i|^2\right)^{1/2} + \left(\sum |b_i|^2\right)^{1/2}.$$

In $\mathbb{R}^2, \mathbb{R}^3$, $\langle x, y\rangle = \|x\|\|y\|\cos\theta$, θ the angle between x and y.

Definition (Orthogonality)

x, y are **orthogonal** (perpendicular) if $\langle x, y\rangle = 0$. $S \subseteq V$ is **orthogonal** if distinct vectors of S are orthogonal, and **orthonormal** if moreover all vectors are **unit vectors** ($\|x\| = 1$). Replacing $x \neq 0$ by $x/\|x\|$ is **normalizing**.

Example 8

$\{(1,1,0),(1,-1,1),(-1,1,2)\}$ is orthogonal in $\mathbb{F}^3$; normalizing gives the orthonormal set $\{\frac{1}{\sqrt{2}}(1,1,0), \frac{1}{\sqrt{3}}(1,-1,1), \frac{1}{\sqrt{6}}(-1,1,2)\}$.

Example 9

In H, $f_n(t) = e^{int}$ ($n \in \mathbb{Z}$) form an orthonormal set: for $m \neq n$, $\langle f_m, f_n\rangle = \frac{1}{2\pi}\int_0^{2\pi} e^{i(m-n)t}\,dt = \frac{1}{2\pi i(m-n)} e^{i(m-n)t}\Big|_0^{2\pi} = 0$, and $\langle f_n, f_n\rangle = 1$.

6.2 The Gram–Schmidt orthogonalization process and orthogonal complements

Motivation. In an orthonormal basis coordinates are just inner products. Gram–Schmidt converts any basis into an orthonormal one, and orthogonal projection onto a subspace gives the closest point. This yields orthogonal complements and best approximations.

Definition (Orthonormal basis)

An **orthonormal basis** is an ordered basis that is orthonormal.

Example 1

The standard ordered basis of $\mathbb{F}^n$ is orthonormal.

Example 2

$\{\frac{1}{\sqrt{5}}(1,2), \frac{1}{\sqrt{5}}(2,-1)\}$ is an orthonormal basis of $\mathbb{R}^2$.

Theorem 6.3

If $S = \{v_1, \ldots, v_k\}$ is orthogonal with nonzero vectors and $y \in \operatorname{span}(S)$, then $y = \sum_{i=1}^{k} \frac{\langle y, v_i\rangle}{\|v_i\|^2} v_i$.

Proof. If $y = \sum a_i v_i$, then $\langle y, v_j\rangle = \sum a_i \langle v_i, v_j\rangle = a_j\|v_j\|^2$. ■

Corollary 1

If S is orthonormal and $y \in \operatorname{span}(S)$, then $y = \sum \langle y, v_i\rangle v_i$.

Proof. $\|v_i\| = 1$ in Theorem 6.3. ∎

Corollary 2

An orthogonal set of nonzero vectors is linearly independent.

Proof. If $\sum a_i v_i = 0$, Theorem 6.3 with $y = 0$ gives $a_i = \langle 0, v_i\rangle / \|v_i\|^2 = 0$. ∎

Example 3

By Corollary 2 the orthonormal set $\{\frac{1}{\sqrt{2}}(1,1,0), \frac{1}{\sqrt{3}}(1,-1,1), \frac{1}{\sqrt{6}}(-1,1,2)\}$ of Example 8 of Section 6.1 is a basis of $\mathbb{R}^3$, and for $x = (2,1,3)$ the coefficients are $\frac{3}{\sqrt{2}}$, $\frac{4}{\sqrt{3}}$, $\frac{5}{\sqrt{6}}$. Also H is infinite-dimensional (Example 9 of Section 6.1).

Theorem 6.4 (Gram–Schmidt)

Let $S = \{w_1, \dots, w_n\}$ be linearly independent. Put $v_1 = w_1$ and

$$v_k = w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j\rangle}{\|v_j\|^2} v_j \qquad (2 \le k \le n). \tag{6.1}$$

Then $S' = \{v_1, \dots, v_n\}$ is an orthogonal set of nonzero vectors with $\operatorname{span}(S') = \operatorname{span}(S)$.

Proof. Induction on n; let $S_k = \{w_1, \dots, w_k\}$, $S'_k = \{v_1, \dots, v_k\}$, and assume S'_{k-1} is orthogonal, of nonzero vectors, with $\operatorname{span} S'_{k-1} = \operatorname{span} S_{k-1}$. If $v_k = 0$, then $w_k \in \operatorname{span} S'_{k-1} = \operatorname{span} S_{k-1}$, contradicting independence. For $i < k$, by (6.1) and orthogonality, $\langle v_k, v_i\rangle = \langle w_k, v_i\rangle - \frac{\langle w_k, v_i\rangle}{\|v_i\|^2}\|v_i\|^2 = 0$. Finally $\operatorname{span} S'_k \subseteq \operatorname{span} S_k$ by (6.1), and both have dimension k (S'_k is independent by Corollary 2), so they are equal. ∎

Example 4

In $\mathbb{R}^4$, $w_1 = (1,0,1,0)$, $w_2 = (1,1,1,1)$, $w_3 = (0,1,2,1)$: $v_1 = w_1$, $v_2 = w_2 - \frac{2}{2}v_1 = (0,1,0,1)$, $v_3 = w_3 - \frac{2}{2}v_1 - \frac{2}{2}v_2 = (-1,0,1,0)$. Normalizing gives the orthonormal basis $\frac{1}{\sqrt{2}}(1,0,1,0)$, $\frac{1}{\sqrt{2}}(0,1,0,1)$, $\frac{1}{\sqrt{2}}(-1,0,1,0)$ of $\operatorname{span}\{w_1, w_2, w_3\}$.

Example 5

On $\mathsf{P}(\mathbb{R})$ with $\langle f, g\rangle = \int_{-1}^{1} f(t)g(t)\,dt$, Gram–Schmidt applied to $1, x, x^2$ gives 1, x, $x^2 - \frac{1}{3}$, and normalizing,

$$u_1 = \tfrac{1}{\sqrt{2}}, \qquad u_2 = \sqrt{\tfrac{3}{2}}\,x, \qquad u_3 = \sqrt{\tfrac{5}{8}}\,(3x^2 - 1).$$

Continuing yields (multiples of) the **Legendre polynomials**.

Theorem 6.5

A nonzero finite-dimensional inner product space V has an orthonormal basis $\beta = \{v_1, \dots, v_n\}$, and $x = \sum_{i=1}^{n} \langle x, v_i\rangle v_i$ for every $x \in V$.

Proof. Apply Theorem 6.4 to any basis and normalize (Corollary 2); the expansion is Corollary 1 of Theorem 6.3. ∎

Example 6

For $f = 1 + 2x + 3x^2$ in the basis of Example 5: $\langle f, u_1\rangle = 2\sqrt{2}$, $\langle f, u_2\rangle = \frac{2\sqrt{6}}{3}$, $\langle f, u_3\rangle = \frac{2\sqrt{10}}{5}$, so $f = 2\sqrt{2}u_1 + \frac{2\sqrt{6}}{3}u_2 + \frac{2\sqrt{10}}{5}u_3$.

Corollary

If $\beta = \{v_1, \ldots, v_n\}$ is an orthonormal basis and $A = [\mathsf{T}]_\beta$, then $A_{ij} = \langle \mathsf{T}(v_j), v_i\rangle$.

Proof. $\mathsf{T}(v_j) = \sum_i \langle \mathsf{T}(v_j), v_i\rangle v_i$ by Theorem 6.5. ■

Definition (Fourier coefficients)

For an orthonormal subset β (possibly infinite), the **Fourier coefficients** of x are the $\langle x, y\rangle$, $y \in \beta$.

Example 7

For $f(t) = t$ in H and $f_n = e^{int}$: $\langle f, f_n\rangle = \frac{-1}{in} = \frac{i}{n}$ for $n \neq 0$ and $\langle f, 1\rangle = \pi$. By Bessel's inequality ($\|x\|^2 \geq \sum |\langle x, v_i\rangle|^2$ for orthonormal $\{v_i\}$, which follows from Theorem 6.6 below) and $\|f\|^2 = \frac{4}{3}\pi^2$, $\frac{4}{3}\pi^2 \geq 2\sum_{n=1}^{k} \frac{1}{n^2} + \pi^2$, so $\sum_{n\geq 1} \frac{1}{n^2} \leq \frac{\pi^2}{6}$ (in fact equality holds).

Definition (Orthogonal complement)

For nonempty $S \subseteq V$, $S^\perp = \{x \in V : \langle x, y\rangle = 0 \text{ for all } y \in S\}$, a subspace.

Example 8

$\{0\}^\perp = V$ and $V^\perp = \{0\}$.

Example 9

In $\mathbb{R}^3$, $\{e_3\}^\perp$ is the xy-plane.

Theorem 6.6

Let W be a finite-dimensional subspace of V and $y \in V$. There are unique $u \in W$, $z \in W^\perp$ with $y = u + z$. If $\{v_1, \ldots, v_k\}$ is an orthonormal basis of W, then $u = \sum_{i=1}^{k} \langle y, v_i\rangle v_i$.

Proof. Put $u = \sum \langle y, v_i\rangle v_i$, $z = y - u$. Then $\langle z, v_j\rangle = \langle y, v_j\rangle - \langle y, v_j\rangle = 0$, so $z \in W^\perp$. If also $y = u' + z'$, then $u - u' = z' - z \in W \cap W^\perp = \{0\}$ (a vector orthogonal to itself is 0). ■

Corollary

u is the unique vector of W closest to y: $\|y - x\| \geq \|y - u\|$ for $x \in W$, with equality iff $x = u$.

Proof. $y - x = z + (u - x)$ with $u - x \in W$ orthogonal to z, so $\|y - x\|^2 = \|z\|^2 + \|u - x\|^2 \geq \|z\|^2 = \|y - u\|^2$ (Pythagoras: $\|a + b\|^2 = \|a\|^2 + \|b\|^2$ if $\langle a, b\rangle = 0$), with equality iff $u = x$. ■

u is the **orthogonal projection** of y on W.

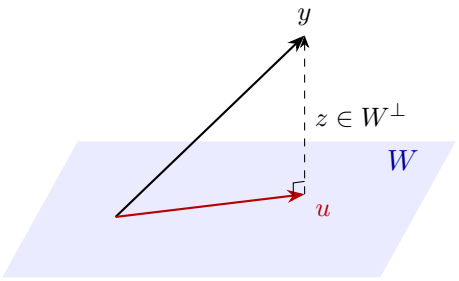


Figure. *Orthogonal projection: $y = u + z$ with $u \in W$, $z \perp W$; u is the point of W closest to y.*

Example 10

In $\mathsf{P}_3(\mathbb{R})$ with $\langle f, g\rangle = \int_{-1}^{1} fg$, the orthogonal projection of x^3 on $\mathsf{P}_2(\mathbb{R})$ is $\sum\langle x^3, u_i\rangle u_i = \frac{\sqrt{6}}{5} \cdot \sqrt{\frac{3}{2}}x = \frac{3}{5}x$ ($\langle x^3, u_1\rangle = \langle x^3, u_3\rangle = 0$).

Theorem 6.7

Let $S = \{v_1, \dots, v_k\}$ be orthonormal in n-dimensional V.
(a) S extends to an orthonormal basis $\{v_1, \dots, v_n\}$ of V.
(b) If $W = \operatorname{span}(S)$, then $\{v_{k+1}, \dots, v_n\}$ is an orthonormal basis of $W^{\perp}$.
(c) For any subspace W, $\dim V = \dim W + \dim W^{\perp}$.

Proof.

(a) Extend S to a basis $\{v_1, \dots, v_k, w_{k+1}, \dots, w_n\}$ and apply Gram–Schmidt. The first k vectors are unchanged (they are already orthonormal), then normalize the rest.

(b) $S_1 = \{v_{k+1}, \dots, v_n\} \subseteq W^{\perp}$ is orthonormal, hence independent. If $x \in W^{\perp}$, then $x = \sum_{i=1}^{n}\langle x, v_i\rangle v_i = \sum_{i>k}\langle x, v_i\rangle v_i$.

(c) Apply (a), (b) to an orthonormal basis of W. ∎

Example 11

For $W = \operatorname{span}\{e_1, e_2\} \subseteq \mathbb{F}^3$, $(a, b, c) \in W^{\perp}$ iff $a = b = 0$, so $W^{\perp} = \operatorname{span}\{e_3\}$.

6.3 The adjoint of a linear operator

Motivation. Every linear functional on a finite-dimensional inner product space is $\langle \cdot, y\rangle$ for a unique y. This produces the adjoint T^*, characterised by $\langle \mathsf{T}x, y\rangle = \langle x, \mathsf{T}^* y\rangle$ and represented by the conjugate transpose. Its first payoffs are least squares and minimal solutions.

Theorem 6.8

If V is finite-dimensional and $g : V \to \mathbb{F}$ is linear, there is a unique $y \in V$ with $g(x) = \langle x, y\rangle$ for all x.

Proof. With an orthonormal basis $\{v_1, \dots, v_n\}$ put $y = \sum_i \overline{g(v_i)} v_i$. Then $h(x) = \langle x, y\rangle$ is linear with $h(v_j) = \langle v_j, y\rangle = g(v_j)$, so $h = g$ (Corollary to Theorem 2.6). Uniqueness: Theorem 6.1(e). ∎

Example 1

$g(a_1, a_2) = 2a_1 + a_2$ on $\mathbb{R}^2$: $y = 2e_1 + e_2 = (2, 1)$.

Theorem 6.9

For a linear operator T on finite-dimensional V there is a unique $\mathsf{T}^* : V \to V$ with $\langle \mathsf{T}(x), y\rangle = \langle x, \mathsf{T}^*(y)\rangle$ for all x, y; T^* is linear.

Proof. For fixed y, $g(x) = \langle \mathsf{T}(x), y\rangle$ is linear, so $g(x) = \langle x, y'\rangle$ for a unique y' (Theorem 6.8); set $\mathsf{T}^*(y) = y'$. Linearity: $\langle x, \mathsf{T}^*(cy_1 + y_2)\rangle = \langle \mathsf{T}x, cy_1 + y_2\rangle = \bar{c}\langle \mathsf{T}x, y_1\rangle + \langle \mathsf{T}x, y_2\rangle = \langle x, c\mathsf{T}^*y_1 + \mathsf{T}^*y_2\rangle$, so $\mathsf{T}^*(cy_1 + y_2) = c\mathsf{T}^*(y_1) + \mathsf{T}^*(y_2)$ by Theorem 6.1(e). Uniqueness: Theorem 6.1(e). ■

T^* is the **adjoint** of T; also $\langle x, \mathsf{T}(y)\rangle = \langle \mathsf{T}^*(x), y\rangle$. (In infinite dimensions an adjoint need not exist.)

Theorem 6.10

If β is an orthonormal basis of finite-dimensional V, then $[\mathsf{T}^*]_\beta = ([\mathsf{T}]_\beta)^*$.

Proof. By the Corollary to Theorem 6.5, $([\mathsf{T}^*]_\beta)_{ij} = \langle \mathsf{T}^*(v_j), v_i\rangle = \overline{\langle v_i, \mathsf{T}^*(v_j)\rangle} = \overline{\langle \mathsf{T}(v_i), v_j\rangle} = \overline{([\mathsf{T}]_\beta)_{ji}}$. ■

Corollary

$\mathsf{L}_{A^*} = (\mathsf{L}_A)^*$ for $A \in \mathrm{M}_{n\times n}(\mathbb{F})$.

Proof. With the standard (orthonormal) basis β, $[(\mathsf{L}_A)^*]_\beta = A^* = [\mathsf{L}_{A^*}]_\beta$. ■

Example 2

$\mathsf{T}(a_1, a_2) = (2ia_1 + 3a_2, a_1 - a_2)$ on $\mathbb{C}^2$: $[\mathsf{T}] = \left(\begin{smallmatrix} 2i & 3 \\ 1 & -1 \end{smallmatrix}\right)$, so $[\mathsf{T}^*] = \left(\begin{smallmatrix} -2i & 1 \\ 3 & -1 \end{smallmatrix}\right)$ and $\mathsf{T}^*(a_1, a_2) = (-2ia_1 + a_2, 3a_1 - a_2)$.

Theorem 6.11

For linear operators T, U on V having adjoints:
(a) $(\mathsf{T} + \mathsf{U})^* = \mathsf{T}^* + \mathsf{U}^*$;
(b) $(c\mathsf{T})^* = \bar{c}\mathsf{T}^*$;
(c) $(\mathsf{TU})^* = \mathsf{U}^*\mathsf{T}^*$;
(d) $\mathsf{T}^{**} = \mathsf{T}$;
(e) $\mathsf{I}^* = \mathsf{I}$.

Proof. Each follows from uniqueness in Theorem 6.9 and Theorem 6.1(e). (a) $\langle (\mathsf{T} + \mathsf{U})x, y\rangle = \langle x, \mathsf{T}^*y + \mathsf{U}^*y\rangle$.

(b) $\langle c\mathsf{T}x, y\rangle = \langle x, \bar{c}\mathsf{T}^*y\rangle$.

(c) $\langle \mathsf{TU}x, y\rangle = \langle \mathsf{U}x, \mathsf{T}^*y\rangle = \langle x, \mathsf{U}^*\mathsf{T}^*y\rangle$.

(d) $\langle \mathsf{T}^*x, y\rangle = \overline{\langle y, \mathsf{T}^*x\rangle} = \overline{\langle \mathsf{T}y, x\rangle} = \langle x, \mathsf{T}y\rangle$.

(e) Clear. ■

Corollary

For $n \times n$ matrices: $(A+B)^* = A^* + B^*$, $(cA)^* = \bar{c}A^*$, $(AB)^* = B^*A^*$, $A^{**} = A$, $I^* = I$.

Proof. Apply Theorem 6.11 to $\mathsf{L}_A, \mathsf{L}_B$ and use the Corollary to Theorem 6.10. (Or compute directly; this works for rectangular matrices too.) ■

6.3.1 Least squares approximation

Given data $(t_1, y_1), \ldots, (t_m, y_m)$, the **least squares line** $y = ct+d$ minimises $E = \sum_i (y_i - ct_i - d)^2 = \|y - Ax\|^2$ with $A = \begin{pmatrix} t_1 & 1 \\ \vdots & \vdots \\ t_m & 1 \end{pmatrix}$, $x = \binom{c}{d}$. Write $\langle \cdot, \cdot \rangle_n$ for the standard inner product on $\mathbb{F}^n$.

Lemma 1

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $x \in \mathbb{F}^n$, $y \in \mathbb{F}^m$: $\langle Ax, y\rangle_m = \langle x, A^*y\rangle_n$.

Proof. $\langle Ax, y\rangle_m = y^*Ax = (A^*y)^*x = \langle x, A^*y\rangle_n$, viewing 1×1 matrices as scalars. ■

Lemma 2

$\mathrm{rank}(A^*A) = \mathrm{rank}(A)$ for $A \in \mathrm{M}_{m\times n}(\mathbb{F})$.

Proof. If $A^*Ax = 0$, then $0 = \langle A^*Ax, x\rangle_n = \langle Ax, Ax\rangle_m$, so $Ax = 0$. Hence $\mathrm{N}(A^*A) = \mathrm{N}(A)$, and the dimension theorem gives equal ranks. ■

Corollary

If $\mathrm{rank}\, A = n$, then A^*A is invertible.

Proof. A^*A is $n \times n$ of rank n by Lemma 2. ■

Theorem 6.12

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $y \in \mathbb{F}^m$, there is $x_0 \in \mathbb{F}^n$ with $(A^*A)x_0 = A^*y$ and $\|Ax_0 - y\| \le \|Ax - y\|$ for all x. If $\mathrm{rank}\, A = n$, then $x_0 = (A^*A)^{-1}A^*y$.

Proof. Let $W = \mathrm{R}(\mathsf{L}_A)$. By the Corollary to Theorem 6.6 there is a closest vector $Ax_0 \in W$ to y, and $Ax_0 - y \in W^\perp$. So $\langle Ax, Ax_0 - y\rangle_m = 0$ for all x, i.e. $\langle x, A^*(Ax_0 - y)\rangle_n = 0$ (Lemma 1), i.e. $A^*Ax_0 = A^*y$. If $\mathrm{rank}\, A = n$, invert A^*A (Corollary to Lemma 2). ■

For data $(1,2), (2,3), (3,5), (4,7)$: $A^*A = \left(\begin{smallmatrix} 30 & 10 \\ 10 & 4 \end{smallmatrix}\right)$, $A^*y = (51, 17)^t$, so $x_0 = \frac{1}{20}\left(\begin{smallmatrix} 4 & -10 \\ -10 & 30 \end{smallmatrix}\right)\binom{51}{17} = \binom{1.7}{0}$. The least squares line is $y = 1.7t$, with error $E = 0.3$. Polynomial fits work the same way with columns t_i^k.

6.3.2 Minimal solutions to systems of linear equations

A solution s of $Ax = b$ is **minimal** if $\|s\| \le \|u\|$ for every solution u.

Theorem 6.13

Let $Ax = b$ be consistent, $A \in \mathrm{M}_{m\times n}(\mathbb{F})$.
(a) There is exactly one minimal solution s, and $s \in \mathrm{R}(\mathsf{L}_{A^*})$.
(b) s is the only solution in $\mathrm{R}(\mathsf{L}_{A^*})$: if $(AA^*)u = b$, then $s = A^*u$.

Proof. Let $W = \mathrm{R}(\mathsf{L}_{A^*})$ and $W' = \mathrm{N}(\mathsf{L}_A)$. By Lemma 1, $W^\perp = W'$: $x \perp A^*y$ for all y iff $\langle Ax, y\rangle = 0$ for all y, iff $Ax = 0$.

(a) Let u be any solution and write $u = s + z$ with $s \in W$, $z \in W^\perp = W'$ (Theorem 6.6). Then $As = Au = b$. Every solution is $s + z'$ with $z' \in W' = W^\perp$ (Theorem 3.9), so $\|s + z'\|^2 = \|s\|^2 + \|z'\|^2 \geq \|s\|^2$, with equality iff $z' = 0$.

(b) If $v \in W$ is a solution, then $v - s \in W \cap W' = W \cap W^\perp = \{0\}$. If $AA^*u = b$, then $A^*u \in W$ is a solution, so it equals s. ∎

Example 3

$x + 2y + z = 4$, $x - y + 2z = -11$, $x + 5y = 19$: $AA^* = \begin{pmatrix} 6 & 1 & 11 \\ 1 & 6 & -4 \\ 11 & -4 & 26 \end{pmatrix}$, and $AA^*u = b$ has the solution $u = (1, -2, 0)$. So the minimal solution is $s = A^*u = (-1, 4, -3)$.

6.4 Normal and self-adjoint operators

Motivation. When does V have an *orthonormal* basis of eigenvectors? Over $\mathbb{C}$ exactly when T commutes with its adjoint (normal); over $\mathbb{R}$ exactly when $\mathsf{T} = \mathsf{T}^*$ (self-adjoint). Schur's triangularization is the stepping stone.

Lemma

If T (on finite-dimensional V) has an eigenvector, so does T^*.

Proof. Let $\mathsf{T}v = \lambda v$, $v \neq 0$. For all x, $0 = \langle(\mathsf{T} - \lambda\mathsf{I})v, x\rangle = \langle v, (\mathsf{T}^* - \bar\lambda\mathsf{I})x\rangle$, so $v \perp \mathrm{R}(\mathsf{T}^* - \bar\lambda\mathsf{I})$, which is therefore not all of V. Hence $\mathsf{T}^* - \bar\lambda\mathsf{I}$ is not onto, so not one-to-one: $\bar\lambda$ is an eigenvalue of T^*. ∎

Theorem 6.14 (Schur)

If the characteristic polynomial of T (on finite-dimensional inner product space V) splits, there is an orthonormal basis β with $[\mathsf{T}]_\beta$ upper triangular.

Proof. Induction on $n = \dim V$; $n = 1$ is clear. The characteristic polynomial splits, so T has an eigenvector, and by the Lemma T^* has a unit eigenvector z, $\mathsf{T}^*z = \lambda z$. $W = \{z\}^\perp$ ($\dim n - 1$) is T-invariant: for $y \in W$, $\langle \mathsf{T}y, z\rangle = \langle y, \mathsf{T}^*z\rangle = \bar\lambda\langle y, z\rangle = 0$. The characteristic polynomial of T_W divides that of T (Theorem 5.21), hence splits. By induction W has an orthonormal basis γ with $[\mathsf{T}_W]_\gamma$ upper triangular. Then $\beta = \gamma \cup \{z\}$ is orthonormal and $[\mathsf{T}]_\beta$ is upper triangular (its first $n - 1$ columns are those of $[\mathsf{T}_W]_\gamma$ padded by 0). ∎

Definition (Normal)

T is **normal** if $\mathsf{T}\mathsf{T}^* = \mathsf{T}^*\mathsf{T}$; A is normal if $AA^* = A^*A$. For orthonormal β, T is normal iff $[\mathsf{T}]_\beta$ is (Theorem 6.10).

Example 1

The rotation T of $\mathbb{R}^2$ by $\theta \in (0,\pi)$, $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, satisfies $AA^* = I = A^*A$, so it is normal but has no eigenvectors: normality does not give an orthonormal eigenbasis over $\mathbb{R}$.

Example 2

A real **skew-symmetric** A ($A^t = -A$) is normal: $AA^t = -A^2 = A^tA$.

Theorem 6.15

Let T be normal on V. Then
(a) $\|\mathsf{T}(x)\| = \|\mathsf{T}^*(x)\|$ for all x;
(b) $\mathsf{T} - c\mathsf{I}$ is normal for every $c \in \mathbb{F}$;
(c) if $\mathsf{T}(x) = \lambda x$, then $\mathsf{T}^*(x) = \bar{\lambda}x$;
(d) eigenvectors for distinct eigenvalues are orthogonal.

Proof.

(a) $\|\mathsf{T}x\|^2 = \langle \mathsf{T}^*\mathsf{T}x, x\rangle = \langle \mathsf{T}\mathsf{T}^*x, x\rangle = \|\mathsf{T}^*x\|^2$.

(b) $(\mathsf{T} - c\mathsf{I})^* = \mathsf{T}^* - \bar{c}\mathsf{I}$ commutes with $\mathsf{T} - c\mathsf{I}$.

(c) Apply (a) to the normal $\mathsf{U} = \mathsf{T} - \lambda\mathsf{I}$: $0 = \|\mathsf{U}x\| = \|\mathsf{U}^*x\| = \|(\mathsf{T}^* - \bar{\lambda}\mathsf{I})x\|$.

(d) If $\mathsf{T}x_1 = \lambda_1 x_1$, $\mathsf{T}x_2 = \lambda_2 x_2$, then by (c) $\lambda_1\langle x_1, x_2\rangle = \langle \mathsf{T}x_1, x_2\rangle = \langle x_1, \mathsf{T}^*x_2\rangle = \langle x_1, \bar{\lambda}_2 x_2\rangle = \lambda_2\langle x_1, x_2\rangle$, so $\langle x_1, x_2\rangle = 0$. ∎

Theorem 6.16

On a finite-dimensional complex inner product space, T is normal iff V has an orthonormal basis of eigenvectors of T.

Proof.

Step 1 (only if). Over $\mathbb{C}$ the characteristic polynomial splits, so Schur gives an orthonormal $\beta = \{v_1, \dots, v_n\}$ with $A = [\mathsf{T}]_\beta$ upper triangular. We show by induction on k that v_k is an eigenvector. v_1 is one (A is upper triangular). If $v_1, \dots, v_{k-1}$ are eigenvectors, $\mathsf{T}v_j = \lambda_j v_j$, then by Theorem 6.15(c), $A_{jk} = \langle \mathsf{T}v_k, v_j\rangle = \langle v_k, \mathsf{T}^*v_j\rangle = \langle v_k, \bar{\lambda}_j v_j\rangle = 0$ for $j < k$. So $\mathsf{T}v_k = A_{kk}v_k$.

Step 2 (if). If β is an orthonormal eigenbasis, $[\mathsf{T}]_\beta = D$ is diagonal, and $[\mathsf{T}^*]_\beta = D^*$ is diagonal too; diagonal matrices commute. ∎

Example 3

In $V = \operatorname{span}\{f_n : n \in \mathbb{Z}\} \subseteq \mathsf{H}$, $\mathsf{T}(f) = f_1 f$ and $\mathsf{U}(f) = f_{-1}f$ satisfy $\mathsf{T}(f_n) = f_{n+1}$, $\mathsf{U}(f_n) = f_{n-1}$, $\mathsf{U} = \mathsf{T}^*$ and $\mathsf{T}\mathsf{T}^* = \mathsf{I} = \mathsf{T}^*\mathsf{T}$. So T is normal, but it has no eigenvectors: if $f = \sum_{i=n}^{m} a_i f_i$ with $a_m \neq 0$ and $\mathsf{T}f = \lambda f$, then $f_{m+1} \in \operatorname{span}\{f_n, \dots, f_m\}$, contradicting independence. So Theorem 6.16 fails in infinite dimensions.

Definition (Self-adjoint)

T is **self-adjoint** (**Hermitian**) if $\mathsf{T} = \mathsf{T}^*$; A is self-adjoint if $A = A^*$. For real matrices this means symmetric.

Lemma

Let T be self-adjoint on finite-dimensional V. (a) Every eigenvalue of T is real.
(b) If V is real, the characteristic polynomial of T splits.

Proof.

(a) T is normal, so $\mathsf{T}x = \lambda x$ gives $\lambda x = \mathsf{T}x = \mathsf{T}^*x = \bar{\lambda}x$ (Theorem 6.15(c)), so $\lambda = \bar{\lambda}$.

(b) Let $A = [\mathsf{T}]_\beta$ for orthonormal β; A is real symmetric, so $A = A^*$ and L_A on $\mathbb{C}^n$ is self-adjoint. Its characteristic polynomial (that of T) splits over $\mathbb{C}$, and by (a) all its zeros are real. ■

Theorem 6.17

On a finite-dimensional real inner product space, T is self-adjoint iff V has an orthonormal basis of eigenvectors of T.

Proof. If T is self-adjoint, its characteristic polynomial splits (Lemma), Schur gives orthonormal β with $A = [\mathsf{T}]_\beta$ upper triangular, and $A^* = [\mathsf{T}^*]_\beta = A$ is lower triangular. So A is diagonal. Conversely, if $[\mathsf{T}]_\beta = D$ is real diagonal for orthonormal β, then $[\mathsf{T}^*]_\beta = D^* = D$, so $\mathsf{T}^* = \mathsf{T}$. ■

Example 4

Complex symmetric matrices need not be normal: $A = \left(\begin{smallmatrix} i & i \\ i & 1 \end{smallmatrix}\right)$ has $(AA^*)_{12} = 1 + i \neq 1 - i = (A^*A)_{12}$.

6.5 Unitary and orthogonal operators and their matrices

Motivation. Operators preserving length are the symmetries of an inner product space: unitary over $\mathbb{C}$, orthogonal over $\mathbb{R}$. They preserve inner products, map orthonormal bases to orthonormal bases, and have matrices with $A^*A = I$. Combined with Schur and the spectral results, they give the matrix forms "unitarily/orthogonally equivalent to diagonal".

Definition (Unitary, orthogonal operator)

A linear T on finite-dimensional V with $\|\mathsf{T}(x)\| = \|x\|$ for all x is **unitary** if $\mathbb{F} = \mathbb{C}$ and **orthogonal** if $\mathbb{F} = \mathbb{R}$. (In infinite dimensions such an operator is called an isometry; if also onto, unitary/orthogonal.)

Example 1

If $h \in \mathsf{H}$ has $|h(x)| = 1$ for all x, then $\mathsf{T}(f) = hf$ satisfies $\|\mathsf{T}f\|^2 = \frac{1}{2\pi}\int |h|^2|f|^2 = \|f\|^2$.

Lemma

If U is self-adjoint on finite-dimensional V and $\langle x, \mathsf{U}(x)\rangle = 0$ for all x, then $\mathsf{U} = \mathsf{T}_0$.

Proof. Choose an orthonormal eigenbasis β (Theorems 6.16, 6.17; U is normal, and self-adjoint over $\mathbb{R}$). For $x \in \beta$ with $\mathsf{U}x = \lambda x$, $0 = \langle x, \lambda x\rangle = \bar{\lambda}$. So U vanishes on β. ■

Theorem 6.18

For linear T on finite-dimensional V the following are equivalent:
(a) $\mathsf{T}\mathsf{T}^* = \mathsf{T}^*\mathsf{T} = \mathsf{I}$;
(b) $\langle \mathsf{T}(x), \mathsf{T}(y)\rangle = \langle x, y\rangle$ for all x, y;
(c) if β is an orthonormal basis, so is $\mathsf{T}(\beta)$;
(d) there is an orthonormal basis β with $\mathsf{T}(\beta)$ orthonormal;
(e) $\|\mathsf{T}(x)\| = \|x\|$ for all x.

Proof.

Step (a)⇒(b). $\langle \mathsf{T}x, \mathsf{T}y\rangle = \langle x, \mathsf{T}^*\mathsf{T}y\rangle = \langle x, y\rangle$.

Step (b)⇒(c). $\langle \mathsf{T}v_i, \mathsf{T}v_j\rangle = \langle v_i, v_j\rangle = \delta_{ij}$, so $\mathsf{T}(\beta)$ is orthonormal with n vectors, hence a basis (Corollary 2 of Theorem 6.3).

Step (c)⇒(d). Orthonormal bases exist (Theorem 6.5).

Step (d)⇒(e). For $x = \sum a_i v_i$, $\|\mathsf{T}x\|^2 = \|\sum a_i \mathsf{T}v_i\|^2 = \sum |a_i|^2 = \|x\|^2$.

Step (e)⇒(a). $\langle x, (\mathsf{I} - \mathsf{T}^*\mathsf{T})x\rangle = \|x\|^2 - \|\mathsf{T}x\|^2 = 0$ for all x, and $\mathsf{I} - \mathsf{T}^*\mathsf{T}$ is self-adjoint, so $\mathsf{T}^*\mathsf{T} = \mathsf{I}$ by the Lemma. Then T is invertible with $\mathsf{T}^{-1} = \mathsf{T}^*$ (finite dimension), so $\mathsf{T}\mathsf{T}^* = \mathsf{I}$ too. ■

So unitary/orthogonal operators preserve inner products, angles and distances, and λ an eigenvalue implies $|\lambda| = 1$ ($\|\mathsf{T}x\| = |\lambda|\|x\|$).

Corollary 1

On a finite-dimensional real inner product space, V has an orthonormal basis of eigenvectors of T with eigenvalues of absolute value 1 iff T is both self-adjoint and orthogonal.

Proof. If T is self-adjoint, it has an orthonormal eigenbasis (Theorem 6.17), and if also orthogonal, eigenvalues have $|\lambda| = 1$. Conversely, such a basis makes T self-adjoint (Theorem 6.17), and $\mathsf{T}(\beta) = \{\pm v_i\}$ is orthonormal, so T is orthogonal by Theorem 6.18(d). ■

Corollary 2

On a finite-dimensional complex inner product space, V has an orthonormal basis of eigenvectors of T with eigenvalues of absolute value 1 iff T is unitary.

Proof. A unitary T is normal ($\mathsf{T}\mathsf{T}^* = \mathsf{I} = \mathsf{T}^*\mathsf{T}$), so Theorem 6.16 gives the basis, with $|\lambda| = 1$. Conversely $\mathsf{T}(\beta) = \{\lambda_i v_i\}$ is orthonormal; apply Theorem 6.18(d). ■

Example 2

The rotation T of $\mathbb{R}^2$ by θ preserves lengths, so it is orthogonal; in the standard basis $[\mathsf{T}] = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and $[\mathsf{T}]^*[\mathsf{T}] = I$.

Definition (Reflection of $\mathbb{R}^2$)

For a line L through 0, T is the **reflection about** L if $\mathsf{T}(x) = x$ for $x \in L$ and $\mathsf{T}(x) = -x$ for $x \in L^\perp$.

Example 3

A reflection is orthogonal: with unit $v_1 \in L$, $v_2 \in L^\perp$, $\mathsf{T}(\{v_1, v_2\}) = \{v_1, -v_2\}$ is orthonormal (Theorem 6.18(d)).

Definition (Orthogonal and unitary matrices)

A square A is **orthogonal** if $A^tA = AA^t = I$ and **unitary** if $A^*A = AA^* = I$. Equivalently, the rows (or columns) form an orthonormal basis of $\mathbb{F}^n$. T is unitary [orthogonal] iff $[\mathsf{T}]_\beta$ is, for orthonormal β.

Example 4

$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ is orthogonal; its rows are orthonormal.

Example 5

For a reflection $\mathsf{T} = \mathsf{L}_A$ about the line L at angle α to the x-axis, $A = Q\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)Q^{-1}$ with $Q = \left(\begin{smallmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{smallmatrix}\right)$, so $A = \left(\begin{smallmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{smallmatrix}\right)$, an orthogonal matrix.

Definition (Unitary/orthogonal equivalence)

A is **unitarily** [**orthogonally**] **equivalent** to D if $A = P^*DP$ for a unitary [orthogonal] P.

Theorem 6.19

A complex $n \times n$ matrix A is normal iff it is unitarily equivalent to a diagonal matrix.

Proof. If A is normal, Theorem 6.16 gives an orthonormal basis β of eigenvectors of L_A; with Q having these columns (Q unitary), $D = Q^{-1}AQ = Q^*AQ$ is diagonal. Conversely, if $A = P^*DP$, then $AA^* = P^*DD^*P = P^*D^*DP = A^*A$. ■

Theorem 6.20

A real $n \times n$ matrix A is symmetric iff it is orthogonally equivalent to a real diagonal matrix.

Proof. As in Theorem 6.19, using Theorem 6.17. Conversely $(P^tDP)^t = P^tDP$. ■

Example 6

$A = \left(\begin{smallmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{smallmatrix}\right)$ has eigenvalues $2, 2, 8$. Gram–Schmidt on the eigenvectors $(-1, 1, 0), (-1, 0, 1)$ of 2, and normalizing $(1, 1, 1)$, gives

$$P = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}, \qquad P^tAP = \operatorname{diag}(2, 2, 8).$$

Theorem 6.21 (Schur, matrix form)

Let the characteristic polynomial of $A \in \mathrm{M}_{n\times n}(\mathbb{F})$ split over $\mathbb{F}$. (a) If $\mathbb{F} = \mathbb{C}$, A is unitarily equivalent to a complex upper triangular matrix.
(b) If $\mathbb{F} = \mathbb{R}$, A is orthogonally equivalent to a real upper triangular matrix.

Proof. Apply Theorem 6.14 to L_A on $\mathbb{F}^n$; the change of coordinate matrix Q from the orthonormal basis β to the standard basis has orthonormal columns, hence is unitary [orthogonal], and $Q^*AQ = [\mathsf{L}_A]_\beta$ is upper triangular. ■

6.5.1 Rigid motions

Definition (Rigid motion; translation)

On a real inner product space V, $f : V \to V$ is a **rigid motion** if $\|f(x) - f(y)\| = \|x - y\|$ for all x, y. A **translation** by v_0 is $g(x) = x + v_0$. Orthogonal operators and translations are rigid motions, and so are their composites.

Theorem 6.22

Every rigid motion f of a finite-dimensional real inner product space is $f = g \circ \mathsf{T}$ for a unique orthogonal operator T and a unique translation g.

Proof.

Step 1 (T preserves norms and inner products). Let $\mathsf{T}(x) = f(x) - f(0)$, a rigid motion with $\mathsf{T}(0) = 0$, so $\|\mathsf{T}x\| = \|x\|$. Expanding $\|\mathsf{T}x - \mathsf{T}y\|^2 = \|x - y\|^2$ gives $\langle \mathsf{T}x, \mathsf{T}y\rangle = \langle x, y\rangle$.

Step 2 (T is linear). For $x, y \in V$, $a \in \mathbb{R}$,

$$\begin{aligned}\|\mathsf{T}(x + ay) - \mathsf{T}x - a\mathsf{T}y\|^2 &= \|\mathsf{T}(x + ay) - \mathsf{T}x\|^2 + a^2\|\mathsf{T}y\|^2 - 2a\langle \mathsf{T}(x + ay) - \mathsf{T}x, \mathsf{T}y\rangle \\ &= a^2\|y\|^2 + a^2\|y\|^2 - 2a(\langle x + ay, y\rangle - \langle x, y\rangle) = 0,\end{aligned}$$

using Step 1 throughout. So T is linear and, preserving norms, orthogonal; $f = g \circ \mathsf{T}$ with g the translation by $f(0)$.

Step 3 (uniqueness). If $\mathsf{T}x + u_0 = \mathsf{U}x + v_0$ for all x, then $x = 0$ gives $u_0 = v_0$, and then $\mathsf{T} = \mathsf{U}$. ■

Theorem 6.23

For an orthogonal operator T on $\mathbb{R}^2$ with $A = [\mathsf{T}]_\beta$ (β standard), exactly one holds:
(a) T is a rotation and $\det A = 1$;
(b) T is a reflection about a line through 0 and $\det A = -1$.

Proof. $\mathsf{T}(e_1)$ is a unit vector, so $\mathsf{T}(e_1) = (\cos\theta, \sin\theta)$ for a unique $\theta \in [0, 2\pi)$, and $\mathsf{T}(e_2)$ is a unit vector orthogonal to it: $(-\sin\theta, \cos\theta)$ or $(\sin\theta, -\cos\theta)$. In the first case A is the rotation matrix, $\det A = 1$. In the second, $A = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$, $\det A = -1$, which is the reflection about the line at angle $\theta/2$ (Example 5). The determinants differ, so exactly one case occurs. ■

Corollary

Every rigid motion of $\mathbb{R}^2$ is a rotation followed by a translation, or a reflection about a line through 0 followed by a translation.

Proof. Theorems 6.22 and 6.23. ∎

Example 7

$A = \frac{1}{\sqrt{5}}\left(\begin{smallmatrix} 1 & 2 \\ 2 & -1 \end{smallmatrix}\right)$ is orthogonal with $\det A = -\frac{1}{5} - \frac{4}{5} = -1$, so L_A is a reflection; its axis is $E_1 = \operatorname{span}\{(1+\sqrt{5}, 2)\}$, the line $y = \frac{\sqrt{5}-1}{2}x$.

6.5.2 Conic sections

A quadratic equation $ax^2 + 2bxy + cy^2 + dx + ey + f = 0$ has quadratic part X^tAX with $X = \binom{x}{y}$, $A = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$. By Theorem 6.20 there is an orthogonal P with $P^tAP = \operatorname{diag}(\lambda_1, \lambda_2)$; replacing P by $P \cdot \operatorname{diag}(1, -1)$ if needed, $\det P = 1$, so P is a rotation. With $X = PX'$, $X^tAX = \lambda_1(x')^2 + \lambda_2(y')^2$: the xy-term disappears by rotating the axes. For example, $2x^2 - 4xy + 5y^2 - 36 = 0$ has $A = \left(\begin{smallmatrix} 2 & -2 \\ -2 & 5 \end{smallmatrix}\right)$ with eigenvalues $1, 6$ and $P = \frac{1}{\sqrt{5}}\left(\begin{smallmatrix} 2 & -1 \\ 1 & 2 \end{smallmatrix}\right)$. It becomes $(x')^2 + 6(y')^2 = 36$, an ellipse.

6.6 Orthogonal projections and the spectral theorem

Motivation. A normal (complex) or self-adjoint (real) operator is a weighted sum of orthogonal projections onto its mutually orthogonal eigenspaces, $\mathsf{T} = \sum \lambda_i \mathsf{T}_i$. This spectral decomposition gives polynomial descriptions of normality and of the projections themselves.

If $V = W_1 \oplus W_2$, the **projection on** W_1 **along** W_2 is $\mathsf{T}(x_1 + x_2) = x_1$; then $\mathrm{R}(\mathsf{T}) = W_1$, $\mathrm{N}(\mathsf{T}) = W_2$, and T is a projection iff $\mathsf{T}^2 = \mathsf{T}$.

Definition (Orthogonal projection)

A projection T on an inner product space is **orthogonal** if $\mathrm{R}(\mathsf{T})^\perp = \mathrm{N}(\mathsf{T})$ and $\mathrm{N}(\mathsf{T})^\perp = \mathrm{R}(\mathsf{T})$ (in finite dimensions one of these suffices). For finite-dimensional W, the orthogonal projection on W is $x \mapsto u$ of Theorem 6.6.

Theorem 6.24

T is an orthogonal projection iff T has an adjoint and $\mathsf{T}^2 = \mathsf{T} = \mathsf{T}^*$.

Proof.

Step 1 (only if). $\mathsf{T}^2 = \mathsf{T}$ as T is a projection. Write $x = x_1 + x_2$, $y = y_1 + y_2$ with $x_1, y_1 \in \mathrm{R}\,\mathsf{T}$, $x_2, y_2 \in \mathrm{N}\,\mathsf{T}$, orthogonal to each other. Then $\langle x, \mathsf{T}y\rangle = \langle x_1 + x_2, y_1\rangle = \langle x_1, y_1\rangle = \langle x_1, y_1 + y_2\rangle = \langle \mathsf{T}x, y\rangle$, so T^* exists and equals T.

Step 2 (if). $\mathsf{T}^2 = \mathsf{T}$ makes T the projection on $\mathrm{R}\,\mathsf{T}$ along $\mathrm{N}\,\mathsf{T}$ (for $x = \mathsf{T}x + (x - \mathsf{T}x)$). If $x \in \mathrm{R}\,\mathsf{T}$, $y \in \mathrm{N}\,\mathsf{T}$, then $\langle x, y\rangle = \langle \mathsf{T}x, y\rangle = \langle x, \mathsf{T}y\rangle = 0$, so $\mathrm{R}\,\mathsf{T} \subseteq \mathrm{N}(\mathsf{T})^\perp$. If $y \in \mathrm{N}(\mathsf{T})^\perp$, then $\|y - \mathsf{T}y\|^2 = \langle y - \mathsf{T}y, y\rangle - \langle y - \mathsf{T}y, \mathsf{T}y\rangle = 0 - \langle \mathsf{T}(y - \mathsf{T}y), y\rangle = 0$, so $y \in \mathrm{R}\,\mathsf{T}$. Hence $\mathrm{N}(\mathsf{T})^\perp = \mathrm{R}\,\mathsf{T}$; the other equality is analogous. ∎

Theorem 6.25 (Spectral theorem)

Let T on finite-dimensional V be normal ($\mathbb{F} = \mathbb{C}$) or self-adjoint ($\mathbb{F} = \mathbb{R}$), with distinct eigenvalues $\lambda_1, \dots, \lambda_k$, eigenspaces W_i, and T_i the orthogonal projection on W_i. Then
(a) $V = W_1 \oplus \cdots \oplus W_k$;
(b) $W_i^\perp = W_i' = \bigoplus_{j \neq i} W_j$;

(c) $\mathsf{T}_i\mathsf{T}_j = \delta_{ij}\mathsf{T}_i$;
(d) $\mathsf{I} = \mathsf{T}_1 + \cdots + \mathsf{T}_k$;
(e) $\mathsf{T} = \lambda_1\mathsf{T}_1 + \cdots + \lambda_k\mathsf{T}_k$.

Proof.

(a) T is diagonalizable (Theorems 6.16, 6.17); apply Theorem 5.11.

(b) Eigenspaces are mutually orthogonal (Theorem 6.15(d)), so $W_i' \subseteq W_i^\perp$; both have dimension $\dim V - \dim W_i$ (by (a) and Theorem 6.7(c)).

(c) For $i \neq j$, $\mathrm{R}\,\mathsf{T}_j = W_j \subseteq W_i^\perp = \mathrm{N}\,\mathsf{T}_i$, so $\mathsf{T}_i\mathsf{T}_j = \mathsf{T}_0$; and $\mathsf{T}_i^2 = \mathsf{T}_i$.

(d) For $x = \sum x_j$, $x_j \in W_j$, $\mathsf{T}_i x = x_i$ by (b), so $\sum \mathsf{T}_i x = x$.

(e) $\mathsf{T}x = \sum \mathsf{T}x_i = \sum \lambda_i x_i = \sum \lambda_i \mathsf{T}_i x$. ∎

$\{\lambda_i\}$ is the **spectrum**, (d) the **resolution of the identity**, and (e) the **spectral decomposition** of T. Note $g(\mathsf{T}) = \sum g(\lambda_i)\mathsf{T}_i$ for any polynomial g, by (c).

Corollary 1

If $\mathbb{F} = \mathbb{C}$, T is normal iff $\mathsf{T}^* = g(\mathsf{T})$ for some polynomial g.

Proof. If T is normal, $\mathsf{T}^* = \sum \bar{\lambda}_i \mathsf{T}_i$ (each T_i is self-adjoint by Theorem 6.24). Lagrange interpolation gives g with $g(\lambda_i) = \bar{\lambda}_i$, so $g(\mathsf{T}) = \sum \bar{\lambda}_i \mathsf{T}_i = \mathsf{T}^*$. Conversely $g(\mathsf{T})$ commutes with T. ∎

Corollary 2

If $\mathbb{F} = \mathbb{C}$, T is unitary iff T is normal and $|\lambda| = 1$ for every eigenvalue.

Proof. Unitary operators are normal with $|\lambda| = 1$. Conversely $\mathsf{T}\mathsf{T}^* = \sum \lambda_i\bar{\lambda}_i\mathsf{T}_i = \sum \mathsf{T}_i = \mathsf{I}$ by (c), (d). ∎

Corollary 3

If $\mathbb{F} = \mathbb{C}$ and T is normal, T is self-adjoint iff every eigenvalue is real.

Proof. $\mathsf{T}^* = \sum \bar{\lambda}_i \mathsf{T}_i$ equals $\mathsf{T} = \sum \lambda_i \mathsf{T}_i$ iff $\bar{\lambda}_i = \lambda_i$ for all i (apply both sides to W_j). The "only if" is also the Lemma before Theorem 6.17. ∎

Corollary 4

In the spectral decomposition $\mathsf{T} = \sum \lambda_i \mathsf{T}_i$, each T_j is a polynomial in T.

Proof. Choose g_j (Lagrange) with $g_j(\lambda_i) = \delta_{ij}$; then $g_j(\mathsf{T}) = \sum g_j(\lambda_i)\mathsf{T}_i = \mathsf{T}_j$. ∎

6.7 The singular value decomposition and the pseudoinverse *

Motivation. Any linear map between inner product spaces, square or not, sends some orthonormal basis to an orthogonal set: $\mathsf{T}v_i = \sigma_i u_i$. This singular value decomposition gives the polar decomposition $A = WP$ and the pseudoinverse $A^\dagger$, which produces minimum-norm least squares solutions.

Theorem 6.26 (Singular value theorem for linear transformations)

Let $\mathsf{T}: V \to W$ (finite-dimensional inner product spaces) have rank r. There are orthonormal bases $\{v_1,\dots,v_n\}$ of V, $\{u_1,\dots,u_m\}$ of W and $\sigma_1 \geq \cdots \geq \sigma_r > 0$ with $\mathsf{T}(v_i) = \sigma_i u_i$ for $i \leq r$ and $\mathsf{T}(v_i) = 0$ for $i > r$. Conversely, given such bases and scalars, the v_i are eigenvectors of $\mathsf{T}^*\mathsf{T}$ with eigenvalues σ_i^2 ($i \leq r$) and 0 ($i > r$); hence the σ_i are uniquely determined by T.

Proof.

Step 1 (existence). The adjoint $\mathsf{T}^*: W \to V$ exists (as in Theorem 6.9), and $\mathsf{T}^*\mathsf{T}$ is self-adjoint with $\operatorname{rank}(\mathsf{T}^*\mathsf{T}) = r$ (Lemma 2 of Section 6.3 in operator form). Take an orthonormal eigenbasis $\{v_i\}$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > 0 = \lambda_{r+1} = \dots$. They are ≥ 0: $\lambda_i = \langle \mathsf{T}^*\mathsf{T}v_i, v_i\rangle = \|\mathsf{T}v_i\|^2$. Put $\sigma_i = \sqrt{\lambda_i}$ and $u_i = \frac{1}{\sigma_i}\mathsf{T}(v_i)$ for $i \leq r$. Then $\langle u_i, u_j\rangle = \frac{1}{\sigma_i\sigma_j}\langle \mathsf{T}^*\mathsf{T}v_i, v_j\rangle = \frac{\lambda_i}{\sigma_i\sigma_j}\delta_{ij} = \delta_{ij}$; extend to an orthonormal basis of W. For $i > r$, $\|\mathsf{T}v_i\|^2 = \lambda_i = 0$.

Step 2 (converse). $\mathsf{T}^*(u_i) = \sum_j \langle \mathsf{T}^*u_i, v_j\rangle v_j = \sum_j \langle u_i, \mathsf{T}v_j\rangle v_j = \sigma_i v_i$ for $i \leq r$ (and 0 for $i > r$), so $\mathsf{T}^*\mathsf{T}v_i = \sigma_i^2 v_i$ for $i \leq r$ and $\mathsf{T}^*\mathsf{T}v_i = 0$ for $i > r$. ∎

The σ_i are the **singular values** of T (extended by zeros up to $\min(m,n)$).

Example 1

$\mathsf{T}(f) = f'$ from $\mathsf{P}_2(\mathbb{R})$ to $\mathsf{P}_1(\mathbb{R})$, $\langle f, g\rangle = \int_{-1}^{1} fg$. In the orthonormal bases $\alpha = \{\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}x, \sqrt{\frac{5}{8}}(3x^2-1)\}$ and $\alpha' = \{\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}x\}$, $A = [\mathsf{T}] = \begin{pmatrix} 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{15} \end{pmatrix}$ and $A^*A = \operatorname{diag}(0, 3, 15)$. So $\sigma_1 = \sqrt{15}$, $\sigma_2 = \sqrt{3}$, with $v_1 = \sqrt{\frac{5}{8}}(3x^2-1)$, $v_2 = \sqrt{\frac{3}{2}}x$, $v_3 = \frac{1}{\sqrt{2}}$, $u_1 = \sqrt{\frac{3}{2}}x$, $u_2 = \frac{1}{\sqrt{2}}$.

Example 2

For invertible T on $\mathbb{R}^2$ with $\mathsf{T}v_i = \sigma_i u_i$, the unit circle $\{x_1v_1 + x_2v_2 : x_1^2 + x_2^2 = 1\}$ maps to $\{y_1u_1 + y_2u_2 : \frac{y_1^2}{\sigma_1^2} + \frac{y_2^2}{\sigma_2^2} = 1\}$, an ellipse with semi-axes σ_1, σ_2 along u_1, u_2.

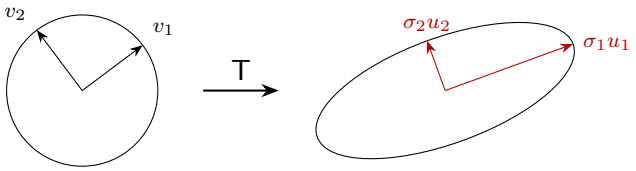


Figure. *Example 2:* T *maps the unit circle onto an ellipse whose semi-axes are* $\sigma_1 u_1$ *and* $\sigma_2 u_2$.

The singular values of a matrix A are those of L_A.

Theorem 6.27 (Singular value decomposition for matrices)

Let $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ have rank r and positive singular values $\sigma_1 \geq \cdots \geq \sigma_r$, and let $\Sigma \in \mathrm{M}_{m\times n}$ have $\Sigma_{ii} = \sigma_i$ ($i \leq r$) and all other entries 0. Then $A = U\Sigma V^*$ for some unitary U ($m \times m$) and V ($n \times n$).

Proof. Apply Theorem 6.26 to L_A; let V, U have columns v_j, u_i. Then AV has columns $Av_j = \sigma_j u_j$ (or 0), which are the columns of $U\Sigma$. So $AV = U\Sigma$ and $A = U\Sigma V^*$. ∎

Such a factorization is a **singular value decomposition** of A.

Example 3

$A = \left(\begin{smallmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{smallmatrix}\right)$: A^*A has eigenvalues $6, 0, 0$ with orthonormal eigenvectors $v_1 = \frac{1}{\sqrt{3}}(1,1,-1)$, $v_2 = \frac{1}{\sqrt{2}}(1,-1,0)$, $v_3 = \frac{1}{\sqrt{6}}(1,1,2)$. So $\sigma_1 = \sqrt{6}$, $u_1 = \frac{1}{\sqrt{6}}Av_1 = \frac{1}{\sqrt{2}}(1,1)$, $u_2 = \frac{1}{\sqrt{2}}(1,-1)$, and $A = U\Sigma V^*$ with $\Sigma = \left(\begin{smallmatrix} \sqrt{6} & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$.

Theorem 6.28 (Polar decomposition)

Every square A is $A = WP$ with W unitary and P positive semidefinite (P self-adjoint with nonnegative eigenvalues). If A is invertible, the representation is unique.

Proof. With $A = U\Sigma V^*$, put $W = UV^*$ (unitary) and $P = V\Sigma V^*$ (self-adjoint, eigenvalues $\sigma_i \geq 0$); then $WP = U\Sigma V^* = A$. If A is invertible and $A = WP = ZQ$, then $P^2 = P^*W^*WP = A^*A = Q^2$. Positive semidefinite square roots are unique (on each eigenspace of A^*A with eigenvalue μ, both P and Q must act as $\sqrt{\mu}$, since P commutes with P^2 and its eigenvalues are ≥ 0). So $P = Q$, P is invertible, and $W = Z = AP^{-1}$. ∎

Example 4

$A = \left(\begin{smallmatrix} 11 & -5 \\ -2 & 10 \end{smallmatrix}\right)$: A^*A has eigenvalues $200, 50$ with $v_1 = \frac{1}{\sqrt{2}}(1,-1)$, $v_2 = \frac{1}{\sqrt{2}}(1,1)$; $u_1 = \frac{1}{5}(4,-3)$, $u_2 = \frac{1}{5}(3,4)$. Then

$$W = UV^* = \frac{1}{5\sqrt{2}}\begin{pmatrix} 7 & -1 \\ 1 & 7 \end{pmatrix}, \qquad P = V\Sigma V^* = \frac{5}{\sqrt{2}}\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}.$$

6.7.1 The pseudoinverse

Definition (Pseudoinverse)

For linear $\mathsf{T} : V \to W$, let $\mathsf{L} : \mathrm{N}(\mathsf{T})^\perp \to \mathrm{R}(\mathsf{T})$ be the restriction of T (an isomorphism). The **pseudoinverse** (Moore–Penrose generalized inverse) $\mathsf{T}^\dagger : W \to V$ is $\mathsf{T}^\dagger(y) = \mathsf{L}^{-1}(y)$ for $y \in \mathrm{R}(\mathsf{T})$ and $\mathsf{T}^\dagger(y) = 0$ for $y \in \mathrm{R}(\mathsf{T})^\perp$. For a matrix, $A^\dagger$ is the matrix with $\mathsf{L}_{A^\dagger} = (\mathsf{L}_A)^\dagger$. If T is invertible, $\mathsf{T}^\dagger = \mathsf{T}^{-1}$.

In the setting of Theorem 6.26, $\mathsf{T}^\dagger(u_i) = \frac{1}{\sigma_i}v_i$ for $i \leq r$ and $\mathsf{T}^\dagger(u_i) = 0$ for $i > r$ ($\{v_1, \ldots, v_r\}$ spans $\mathrm{N}(\mathsf{T})^\perp$ and $\{u_1, \ldots, u_r\}$ spans $\mathrm{R}\,\mathsf{T}$).

Example 5

For $\mathsf{T} = d/dx$ of Example 1: $\mathsf{T}^\dagger(u_1) = \frac{1}{\sqrt{15}}v_1$, $\mathsf{T}^\dagger(u_2) = \frac{1}{\sqrt{3}}v_2$; e.g. $\mathsf{T}^\dagger(1) = \mathsf{T}^\dagger(\sqrt{2}u_2) = \sqrt{\frac{2}{3}}\sqrt{\frac{3}{2}}x = x$.

Theorem 6.29

If $A = U\Sigma V^*$ is a singular value decomposition and $\Sigma^\dagger \in \mathrm{M}_{n\times m}$ has $\Sigma^\dagger_{ii} = \frac{1}{\sigma_i}$ ($i \leq r$) and other entries 0, then $A^\dagger = V\Sigma^\dagger U^*$, and this is a singular value decomposition of $A^\dagger$.

Proof. $V\Sigma^\dagger U^* u_i = V\Sigma^\dagger e_i = \frac{1}{\sigma_i}v_i$ for $i \leq r$ and 0 for $i > r$, matching $(\mathsf{L}_A)^\dagger$ on the basis $\{u_i\}$. The factors are unitary and $\Sigma^\dagger$ has the required form (with values $\frac{1}{\sigma_r} \geq \cdots \geq \frac{1}{\sigma_1}$ after reordering). ∎

Example 6

For A of Example 3, $A^\dagger = V\Sigma^\dagger U^* = \frac{1}{\sqrt{6}} v_1 u_1^* = \frac{1}{6}\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{pmatrix}$.

Lemma

For linear $\mathsf{T} : V \to W$ (finite-dimensional): (a) $\mathsf{T}^\dagger\mathsf{T}$ is the orthogonal projection of V on $\mathrm{N}(\mathsf{T})^\perp$;
(b) $\mathsf{T}\mathsf{T}^\dagger$ is the orthogonal projection of W on $\mathrm{R}(\mathsf{T})$.

Proof.

(a) For $x \in \mathrm{N}(\mathsf{T})^\perp$, $\mathsf{T}^\dagger\mathsf{T}x = \mathsf{L}^{-1}\mathsf{L}x = x$; for $x \in \mathrm{N}(\mathsf{T})$, $\mathsf{T}^\dagger\mathsf{T}x = 0$.

(b) For $y \in \mathrm{R}\,\mathsf{T}$, $\mathsf{T}\mathsf{T}^\dagger y = \mathsf{L}\mathsf{L}^{-1}y = y$; for $y \in \mathrm{R}(\mathsf{T})^\perp$, $\mathsf{T}^\dagger y = 0$. ■

Theorem 6.30

For $Ax = b$ ($A \in \mathrm{M}_{m\times n}(\mathbb{F})$) let $z = A^\dagger b$.
(a) If $Ax = b$ is consistent, z is its unique solution of minimal norm.
(b) If $Ax = b$ is inconsistent, z is the unique best approximation of minimal norm: $\|Az - b\| \le \|Ay - b\|$ for all y, with equality iff $Az = Ay$, and then $\|z\| \le \|y\|$, with equality iff $z = y$.

Proof.

(a) $Az = AA^\dagger b$ is the orthogonal projection of b on $\mathrm{R}\,\mathsf{L}_A$ (Lemma), which is b itself. $z \in \mathrm{R}(\mathsf{L}_{A^\dagger}) = \mathrm{N}(\mathsf{L}_A)^\perp = \mathrm{R}(\mathsf{L}_{A^*})$, so z is the minimal solution by Theorem 6.13.

(b) Az is the projection of b on $\mathrm{R}\,\mathsf{L}_A$, hence the closest point (Corollary to Theorem 6.6): $\|Az - b\| \le \|Ay - b\|$, with equality iff $Ay = Az$. Among such y (the solutions of $Ax = Az$), $z \in \mathrm{N}(\mathsf{L}_A)^\perp$ has minimal norm by (a). ■

Example 7

For $x_1 + x_2 - x_3 = 1$, $x_1 + x_2 - x_3 = 1$, with A of Example 3: $z = A^\dagger(1,1)^t = \frac{1}{3}(1,1,-1)$ is the minimal solution. For the inconsistent right side $(1,2)^t$, $z = A^\dagger(1,2)^t = \frac{1}{2}(1,1,-1)$ is the minimal-norm best approximation.

6.8 Bilinear and quadratic forms *

Motivation. A bilinear form is represented by a matrix that changes by congruence Q^tAQ rather than by similarity. Symmetric forms can always be diagonalized, over $\mathbb{R}$ even by an orthonormal basis. This simplifies quadric surfaces, gives the second derivative test, and leads to Sylvester's law: the numbers of positive and negative squares are invariants.

Definition (Bilinear form)

$H : V \times V \to \mathbb{F}$ is a **bilinear form** if it is linear in each variable when the other is fixed: $H(ax_1 + x_2, y) = aH(x_1, y) + H(x_2, y)$ and $H(x, ay_1 + y_2) = aH(x, y_1) + H(x, y_2)$. $\mathcal{B}(V)$ is the set of bilinear forms on V.

Example 1

$H(a,b) = 2a_1b_1 + 3a_1b_2 + 4a_2b_1 - a_2b_2$ on $\mathbb{R}^2$ equals x^tAy with $A = \left(\begin{smallmatrix} 2 & 3 \\ 4 & -1 \end{smallmatrix}\right)$, hence is bilinear.

Example 2

For any $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, $H(x,y) = x^tAy$ is a bilinear form on $\mathbb{F}^n$. An inner product on a real space is bilinear; on a complex space it is not (conjugate linear in y).

Properties: $H(0,x) = H(x,0) = 0$; for fixed x, $L_x(y) = H(x,y)$ and $R_x(y) = H(y,x)$ are linear.

Theorem 6.31

Sums and scalar multiples of bilinear forms are bilinear forms, and $\mathcal{B}(V)$ is a vector space under $(H_1 + H_2)(x,y) = H_1(x,y) + H_2(x,y)$, $(aH)(x,y) = aH(x,y)$.

Proof. Bilinearity is preserved by these pointwise operations, and $\mathcal{B}(V)$ is a subspace of $\mathcal{F}(V \times V, \mathbb{F})$ containing the zero form. ■

Definition (Matrix of a bilinear form)

For an ordered basis $\beta = \{v_1, \ldots, v_n\}$, $\psi_\beta(H) = A$ with $A_{ij} = H(v_i, v_j)$ is the **matrix representation** of H.

Example 3

For H of Example 1 and $\beta = \{(1,1),(1,-1)\}$: $\psi_\beta(H) = \left(\begin{smallmatrix} 8 & 4 \\ 2 & -6 \end{smallmatrix}\right)$ (e.g. $B_{11} = 2+3+4-1$); in the standard basis $\psi(H) = \left(\begin{smallmatrix} 2 & 3 \\ 4 & -1 \end{smallmatrix}\right)$.

Theorem 6.32

$\psi_\beta : \mathcal{B}(V) \to \mathrm{M}_{n\times n}(\mathbb{F})$ is an isomorphism ($\dim V = n$).

Proof. ψ_β is linear. If $\psi_\beta(H) = O$, then $H(v_i, \cdot) = 0$ on β, so $H(v_i, y) = 0$ for all y; thus $H(\cdot, y) = 0$ on β, hence $H = 0$. Onto: given A, $H(x,y) = [\phi_\beta(x)]^t A[\phi_\beta(y)]$ is bilinear with $H(v_i, v_j) = e_i^t A e_j = A_{ij}$. ■

Corollary 1

$\dim \mathcal{B}(V) = n^2$.

Proof. Theorems 6.32 and 2.19. ■

Corollary 2

$\psi_\beta(H) = A$ iff $H(x,y) = [\phi_\beta(x)]^t A[\phi_\beta(y)]$ for all x, y.

Proof. The form on the right has matrix A (proof of Theorem 6.32); apply injectivity of ψ_β. ■

Corollary 3

For every $H \in \mathcal{B}(\mathbb{F}^n)$ there is a unique A with $H(x,y) = x^t A y$, namely $A = \psi_\beta(H)$ for the standard β.

Proof. Corollary 2 with $\phi_\beta = \mathsf{I}$. ■

Example 4

$H(a,b) = \det\begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix} = a_1 b_2 - a_2 b_1$ has $A_{ij} = H(e_i, e_j)$, i.e. $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.

Definition (Congruence)

B is **congruent** to A if $B = Q^t A Q$ for some invertible Q; congruence is an equivalence relation.

Theorem 6.33

If Q changes γ-coordinates into β-coordinates, then $\psi_\gamma(H) = Q^t \psi_\beta(H) Q$ for every $H \in \mathcal{B}(V)$.

Proof. $w_j = \sum_k Q_{kj} v_k$, so $H(w_i, w_j) = \sum_{k,l} Q_{ki} H(v_k, v_l) Q_{lj} = (Q^t \psi_\beta(H) Q)_{ij}$. ■

Corollary

If $B = Q^t \psi_\beta(H) Q$ with Q invertible, then $B = \psi_\gamma(H)$ for the ordered basis γ with $w_j = \sum_i Q_{ij} v_i$ (so Q changes γ- into β-coordinates).

Proof. γ is a basis since Q is invertible; apply Theorem 6.33. ■

Definition (Symmetric, diagonalizable forms)

H is **symmetric** if $H(x,y) = H(y,x)$; it is **diagonalizable** if $\psi_\beta(H)$ is diagonal for some β.

Theorem 6.34

H is symmetric iff $\psi_\beta(H)$ is symmetric.

Proof. If H is symmetric, $A_{ij} = H(v_i, v_j) = H(v_j, v_i) = A_{ji}$. Conversely, if A is symmetric, $\bar{H}(x,y) = H(y,x)$ is bilinear with $\psi_\beta(\bar{H}) = A^t = A$, so $\bar{H} = H$. ■

Corollary

A diagonalizable bilinear form is symmetric.

Proof. Diagonal matrices are symmetric; apply Theorem 6.34. ■

Example 5

Over $\mathbb{F} = \mathbb{Z}_2$, $H(a,b) = a_1 b_2 + a_2 b_1$ on $\mathbb{F}^2$ is symmetric with $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ of rank 2, but it is not diagonalizable: a congruent diagonal matrix would be I (the only nonzero scalar is 1). But

$H(x,x) = 2a_1a_2 = 0$ for all x, whereas a form with diagonal matrix I has $H(e_1,e_1) = 1$.

Lemma

If $H \neq 0$ is a symmetric bilinear form on V over a field of characteristic $\neq 2$, then $H(x,x) \neq 0$ for some x.

Proof. Take u, v with $H(u,v) \neq 0$. If $H(u,u) = H(v,v) = 0$, then $H(u+v, u+v) = 2H(u,v) \neq 0$. ■

Theorem 6.35

Over a field of characteristic $\neq 2$, every symmetric bilinear form on finite-dimensional V is diagonalizable.

Proof. Induction on $n = \dim V$; $n = 1$ is trivial, and $H = 0$ is diagonal. Otherwise choose x with $H(x,x) \neq 0$ (Lemma). $L_x(y) = H(x,y)$ is a nonzero functional, so $\dim \mathrm{N}(L_x) = n-1$, and $x \notin \mathrm{N}(L_x)$. The restriction of H to $\mathrm{N}(L_x)$ is symmetric, so by induction $\mathrm{N}(L_x)$ has a basis $\{v_1, \ldots, v_{n-1}\}$ with $H(v_i, v_j) = 0$ $(i \neq j)$. With $v_n = x$, $H(v_i, v_n) = H(x, v_i) = 0$ for $i < n$, and $\{v_1, \ldots, v_n\}$ is a basis diagonalizing H. ■

Corollary

Over a field of characteristic $\neq 2$, every symmetric matrix is congruent to a diagonal matrix.

Proof. Apply Theorem 6.35 to $H(x,y) = x^tAy$ on $\mathbb{F}^n$ and use Theorem 6.33. ■

Computing. Write $Q = E_1 \cdots E_k$ (elementary). Then $D = Q^tAQ = E_k^t \cdots E_1^t A E_1 \cdots E_k$: perform each elementary column operation followed by the same row operation.

Example 6

$A = \begin{pmatrix} 1 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & 1 \end{pmatrix}$: add column 1 to column 2 (and row 1 to row 2); subtract 3×column 1 from column 3 (and the rows likewise); subtract 4×column 2 from column 3 (and the rows likewise). With

$$Q = \begin{pmatrix} 1 & 1 & -7 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{pmatrix}, \qquad Q^tAQ = \mathrm{diag}(1, 1, -24).$$

Definition (Quadratic form)

$K : V \to \mathbb{F}$ is a **quadratic form** if $K(x) = H(x,x)$ for a symmetric $H \in \mathcal{B}(V)$. In characteristic $\neq 2$, H is recovered: $H(x,y) = \frac{1}{2}(K(x+y) - K(x) - K(y))$.

Example 7

A homogeneous polynomial of degree 2, $f(t_1, \ldots, t_n) = \sum_{i \leq j} a_{ij} t_i t_j$, is a quadratic form on $\mathbb{R}^n$: $f(x) = x^tAx$ with $A_{ii} = a_{ii}$ and $A_{ij} = A_{ji} = \frac{1}{2}a_{ij}$ for $i < j$.

Theorem 6.36

For a symmetric bilinear form H on a finite-dimensional real inner product space there is an orthonormal basis β with $\psi_\beta(H)$ diagonal.

Proof. Let γ be orthonormal and $A = \psi_\gamma(H)$, symmetric. By Theorem 6.20, $Q^tAQ = D$ for an orthogonal Q. Let β be defined by Q as in the Corollary to Theorem 6.33; its vectors $w_j = \sum_i Q_{ij}v_i$ are orthonormal (the columns of Q are orthonormal), and $\psi_\beta(H) = D$. ∎

Corollary

For a quadratic form K on a finite-dimensional real inner product space there are an orthonormal basis $\{v_1, \dots, v_n\}$ and scalars λ_i with $K(\sum s_i v_i) = \sum \lambda_i s_i^2$. (The λ_i are the eigenvalues of the matrix of K in any orthonormal basis.)

Proof. Theorem 6.36: $K(x) = H(x,x) = \sum_{i,j} s_i s_j H(v_i, v_j) = \sum \lambda_i s_i^2$. ∎

Example 8

$f(t_1, t_2) = 5t_1^2 + 2t_2^2 + 4t_1t_2$ has $A = \left(\begin{smallmatrix} 5 & 2 \\ 2 & 2 \end{smallmatrix}\right)$ with eigenvalues $6, 1$ and unit eigenvectors $x_1 = \frac{1}{\sqrt{5}}(2,1)$, $x_2 = \frac{1}{\sqrt{5}}(-1,2)$. In coordinates $(t_1,t_2) = s_1x_1 + s_2x_2$, $f = 6s_1^2 + s_2^2$.

Example 9

The surface $2t_1^2 + 6t_1t_2 + 5t_2^2 - 2t_2t_3 + 2t_3^2 + 3t_1 - 2t_2 - t_3 + 14 = 0$: the quadratic part has matrix $\left(\begin{smallmatrix} 2 & 3 & 0 \\ 3 & 5 & -1 \\ 0 & -1 & 2 \end{smallmatrix}\right)$ with eigenvalues $2, 7, 0$ and unit eigenvectors $v_1 = \frac{1}{\sqrt{10}}(1,0,3)$, $v_2 = \frac{1}{\sqrt{35}}(-3,-5,1)$, $v_3 = \frac{1}{\sqrt{14}}(-3,2,1)$. In the coordinates s_i relative to $\{v_1, v_2, v_3\}$ the linear part is $-\sqrt{14}\, s_3$, so the equation becomes $s_3 = \frac{2s_1^2 + 7s_2^2}{\sqrt{14}} + \sqrt{14}$: an elliptic paraboloid.

6.8.1 The second derivative test

For $f : \mathbb{R}^n \to \mathbb{R}$ with continuous third partial derivatives, p is a **critical point** if all $\frac{\partial f}{\partial t_i}(p) = 0$; the **Hessian** $A(p)$ has entries $\frac{\partial^2 f}{\partial t_i \partial t_j}(p)$ and is symmetric.

Theorem 6.37 (Second derivative test)

Let p be a critical point of f with Hessian $A(p)$.
(a) If all eigenvalues of $A(p)$ are positive, f has a local minimum at p.
(b) If all are negative, f has a local maximum at p.
(c) If $A(p)$ has a positive and a negative eigenvalue, p is a saddle point (no local extremum).
(d) If rank $A(p) < n$ and $A(p)$ does not have eigenvalues of both signs, the test is inconclusive.

Proof. Translating, assume $p = 0$ and $f(0) = 0$. By Taylor's theorem, $f(x) = \frac{1}{2}x^tA(0)x + S(x)$ with $S(x)/\|x\|^2 \to 0$ as $x \to 0$. Choose an orthonormal eigenbasis $\{v_i\}$ of $A(0)$ (Theorem 6.20) with eigenvalues λ_i; for $x = \sum s_i v_i$, $\frac{1}{2}x^tA(0)x = \frac{1}{2}\sum \lambda_i s_i^2$.

(a) Let $\lambda = \min \lambda_i > 0$. Choose $\varepsilon > 0$ with $|S(x)| < \frac{\lambda}{4}\|x\|^2$ for $0 < \|x\| < \varepsilon$. Then $f(x) \geq \frac{\lambda}{2}\|x\|^2 - \frac{\lambda}{4}\|x\|^2 > 0 = f(0)$.

(b) Apply (a) to $-f$.

(c) If $\lambda_1 > 0 > \lambda_2$, then for small $s \neq 0$, $f(sv_1) = \frac{1}{2}\lambda_1 s^2 + S(sv_1) > 0$ and $f(sv_2) = \frac{1}{2}\lambda_2 s^2 + S(sv_2) < 0$, so 0 is not a local extremum.

(d) $f = t_1^2 + t_2^4$, $f = t_1^2 - t_2^4$ and $f = t_1^2 + t_2^3$ all have Hessian $\operatorname{diag}(2, 0)$ at 0, and they have, respectively, a minimum, no extremum, and no extremum there. ■

6.8.2 Sylvester's law of inertia

Theorem 6.38 (Sylvester's law of inertia)

For a symmetric bilinear form H on a finite-dimensional real vector space, the numbers of positive and of negative diagonal entries are the same in every diagonal matrix representation of H.

Proof. The number of nonzero diagonal entries is rank of the matrix, which is invariant under congruence (Theorem 3.4); so it suffices to treat the positive entries. Let $\beta = \{v_i\}$ and $\gamma = \{w_i\}$ diagonalize H, with $H(v_i, v_i) > 0$ exactly for $i \leq p$ and $H(w_i, w_i) > 0$ exactly for $i \leq q$, and suppose $p < q$. Let $L(x) = (H(x, v_1), \ldots, H(x, v_p), H(x, w_{q+1}), \ldots, H(x, w_r))$, $r =$ rank, a linear map $V \to \mathbb{R}^{p+r-q}$. Since $p + r - q < r$, nullity $L > n - r$, so some $x_0 \notin \operatorname{span}\{v_{r+1}, \ldots, v_n\}$ has $L(x_0) = 0$. Write $x_0 = \sum a_j v_j = \sum b_j w_j$. Then $H(x_0, v_i) = a_i H(v_i, v_i) = 0$ gives $a_i = 0$ for $i \leq p$, and $b_i = 0$ for $q < i \leq r$. Hence

$$H(x_0, x_0) = \sum_{j=p+1}^{r} a_j^2 H(v_j, v_j) \leq 0 \quad \text{and} \quad H(x_0, x_0) = \sum_{j=1}^{q} b_j^2 H(w_j, w_j) \geq 0,$$

so both vanish. The first gives $a_j = 0$ for $p < j \leq r$, so $x_0 \in \operatorname{span}\{v_{r+1}, \ldots, v_n\}$, a contradiction. By symmetry $p = q$. ■

Definition (Index, signature)

The number of positive diagonal entries in a diagonal representation is the **index**; (positive) − (negative) is the **signature**. Rank, index and signature are the **invariants**; any two determine the third. The same terms apply to real symmetric matrices via congruence.

Example 10

The form of Example 9 has a diagonal representation $\operatorname{diag}(2, 7, 0)$: rank 2, index 2, signature 2.

Example 11

$K(x, y) = x^2 - y^2$ has matrix $\operatorname{diag}(1, -1)$, and in the basis $\{(1, 1), (1, -1)\}$ the matrix $\left(\begin{smallmatrix} 0 & 2 \\ 2 & 0 \end{smallmatrix}\right)$ (not diagonal); rank 2, index 1, signature 0.

Corollary 1 (Sylvester's law for matrices)

For a real symmetric A, the numbers of positive and negative diagonal entries of a diagonal matrix congruent to A do not depend on the choice of the diagonal matrix.

Proof. Apply Theorem 6.38 to $H(x, y) = x^t A y$ and use the Corollary to Theorem 6.33. ■

Corollary 2

Two real symmetric $n \times n$ matrices are congruent iff they have the same invariants.

Proof. Congruent matrices are congruent to the same diagonal matrices, so they have the same invariants (Corollary 1). Conversely, a real symmetric matrix with index p and rank r is congruent to some $D = \operatorname{diag}(d_1, \ldots, d_n)$ with $d_i > 0$ $(i \le p)$, $d_i < 0$ $(p < i \le r)$, $d_i = 0$ $(i > r)$ (reorder by a permutation Q). With $Q = \operatorname{diag}(|d_i|^{-1/2} \text{ or } 1)$, $Q^t D Q = J_{pr} = \operatorname{diag}(I_p, -I_{r-p}, O)$. Matrices with the same invariants are both congruent to J_{pr}. ■

Corollary 3

A real symmetric $n \times n$ matrix has index p and rank r iff it is congruent to J_{pr}.

Proof. The proof of Corollary 2, and the fact that J_{pr} has index p and rank r. ■

Example 12

A of Example 6 is congruent to $\operatorname{diag}(1, 1, -24)$: rank 3, index 2. $B = \left(\begin{smallmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$ has eigenvalues $2, -1, -1$: rank 3, index 1. $C = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & -1 \end{smallmatrix}\right)$ has eigenvalues $1, \frac{1 \pm \sqrt{13}}{2}$: rank 3, index 2. So A and C are congruent, and B is congruent to neither (eigenvalues give a diagonal congruent matrix by Theorem 6.20).

6.9 Einstein's special theory of relativity *

Motivation. If the speed of light is the same in every inertial frame, the coordinate change between two frames in relative motion must preserve the light cone $x^2 + y^2 + z^2 - t^2 = 0$. A few axioms then force the Lorentz transformation, and time dilation follows.

Frames S, S' have parallel axes, S' moves along the x-axis of S with velocity $0 < v < 1$ (units: light seconds and seconds, so the speed of light is 1), and the clocks C, C' read 0 when the origins coincide. An event has **space–time coordinates** $(x, y, z, t)^t$ relative to S, C and $(x', y', z', t')^t$ relative to S', C'; $\mathsf{T}_v : \mathbb{R}^4 \to \mathbb{R}^4$ maps the former to the latter.

Axioms. (R1) Light has speed 1 in either frame. (R2) T_v is an isomorphism. (R3) $y' = y$ and $z' = z$. (R4) x', t' do not depend on y, z. (R5) The origin of S moves along the x'-axis with velocity $-v$ as measured in S'. T_v is the **Lorentz transformation** in direction x.

Theorem 6.39

On $\mathbb{R}^4$ (standard inner product):
(a) $\mathsf{T}_v(e_i) = e_i$ for $i = 2, 3$;
(b) $\operatorname{span}\{e_2, e_3\}$ is T_v-invariant;
(c) $\operatorname{span}\{e_1, e_4\}$ is T_v-invariant;
(d) both are T_v^*-invariant;
(e) $\mathsf{T}_v^*(e_i) = e_i$ for $i = 2, 3$.

Proof.

(a) $\mathsf{T}_v(0) = 0$, so by (R4) the first and fourth coordinates of $\mathsf{T}_v(0, a, b, 0)^t$ are 0 for all a, b, and by (R3) $\mathsf{T}_v(e_2) = e_2$, $\mathsf{T}_v(e_3) = e_3$.

(b) Follows from (a).

(c) [Supplement] By (R3) the second and third coordinates of $\mathsf{T}_v(x,0,0,t)^t$ are 0.

(d) [Supplement] $W = \operatorname{span}\{e_2, e_3\}$ and $W^\perp = \operatorname{span}\{e_1, e_4\}$ are both T_v-invariant. If $\mathsf{T}_v(W) \subseteq W$, then for $x \in W^\perp$, $w \in W$: $\langle \mathsf{T}_v^* x, w\rangle = \langle x, \mathsf{T}_v w\rangle = 0$, so $\mathsf{T}_v^*(W^\perp) \subseteq W^\perp$; symmetrically for W.

(e) For $j \neq 2$, $\langle \mathsf{T}_v^* e_2, e_j\rangle = \langle e_2, \mathsf{T}_v e_j\rangle = 0$ by (a), (c); so $\mathsf{T}_v^* e_2 = k e_2$ with $k = \langle \mathsf{T}_v^* e_2, e_2\rangle = \langle e_2, \mathsf{T}_v e_2\rangle = 1$. Similarly for e_3. ■

A light flash at the common origin at time 0 is seen at (x,y,z) at time $t \geq 0$ iff $x^2+y^2+z^2-t^2 = 0$, and by (R1) the same holds in primed coordinates. Let $A = \operatorname{diag}(1,1,1,-1)$, so $\langle \mathsf{L}_A(w), w\rangle = x^2+y^2+z^2-t^2$.

Theorem 6.40

If $\langle \mathsf{L}_A(w), w\rangle = 0$, then $\langle \mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v(w), w\rangle = 0$.

Proof. If $t \geq 0$, w is an event of the flash, so its primed coordinates $\mathsf{T}_v w$ satisfy $\langle \mathsf{L}_A \mathsf{T}_v w, \mathsf{T}_v w\rangle = 0$, which is $\langle \mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v w, w\rangle$. If $t < 0$, apply this to $-w$. ■

Let $w_1 = (1,0,0,1)^t$, $w_2 = (1,0,0,-1)^t$, an orthogonal basis of $\operatorname{span}\{e_1, e_4\}$, which is invariant under $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v$ by Theorem 6.39.

Theorem 6.41

There are nonzero scalars a, b with (a) $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v(w_1) = a w_2$;
(b) $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v(w_2) = b w_1$.

Proof.

(a) $\langle \mathsf{L}_A w_1, w_1\rangle = 0$, so by Theorem 6.40 $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v(w_1) \perp w_1$, and it lies in $\operatorname{span}\{w_1, w_2\}$; hence it is $a w_2$. Since $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v$ is invertible, $a \neq 0$.

(b) Same, with $\langle \mathsf{L}_A w_2, w_2\rangle = 0$. ■

Corollary

With $B_v = [\mathsf{T}_v]_\beta$ (β standard): (a) $B_v^* A B_v = A$;
(b) $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v = \mathsf{L}_A$.

Proof [Supplement] (left as an exercise in the book). $M = \mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v$ is self-adjoint, so $2a = \langle M w_1, w_2\rangle = \langle w_1, M w_2\rangle = 2b$, i.e. $a = b$. Since $A w_1 = w_2$ and $A w_2 = w_1$, $M = a \mathsf{L}_A$ on $\operatorname{span}\{e_1, e_4\}$, and $M = \mathsf{L}_A$ on $\operatorname{span}\{e_2, e_3\}$ by Theorem 6.39. So $\langle A \mathsf{T}_v w, \mathsf{T}_v w\rangle = a\langle Aw, w\rangle$ for $w \in \operatorname{span}\{e_1, e_4\}$. The axioms as stated do not force $a = 1$: rescaling T_v on $\operatorname{span}\{e_1, e_4\}$ by $\sqrt{a}$ preserves (R1)–(R5). We therefore add the physically natural *reciprocity* assumption: $\mathsf{T}_{-v} = \mathsf{T}_v^{-1}$, and $\mathsf{T}_{-v} = R \mathsf{T}_v R$ for the reflection $R = \operatorname{diag}(-1,1,1,1)$ (reversing the x-axes). Then $a(-v) = a(v)$ (since R preserves $\langle A\cdot,\cdot\rangle$), and $a(v)a(-v) = 1$ (since $\mathsf{T}_{-v}\mathsf{T}_v = \mathsf{I}$). So $a^2 = 1$. Moreover $a > 0$: by (R5), $\mathsf{T}_v(e_4) = (-vt'', 0, 0, t'')^t$ with $t'' > 0$, and then $a \cdot (-1) = a\langle Ae_4, e_4\rangle = (v^2-1)t''^2 < 0$. Hence $a = 1$, which is (b), and (a) is its matrix form. ■

One second (on C) after the origins coincide, the origin of S' has S-coordinates $(v,0,0,1)^t$ and S'-coordinates $(0,0,0,t')^t$ with $t' > 0$. By the Corollary, $v^2 - 1 = \langle \mathsf{L}_A(v,0,0,1)^t, (v,0,0,1)^t\rangle = \langle \mathsf{L}_A(0,0,0,t')^t, (0,0,0,t')^t\rangle = -t'^2$, so $t' = \sqrt{1-v^2}$ and $\mathsf{T}_v(v,0,0,1)^t = (0,0,0,\sqrt{1-v^2})^t$. Likewise (R5) gives $\mathsf{T}_v(e_4) = (-vt'', 0, 0, t'')^t$ with $t'' = 1/\sqrt{1-v^2}$.

Theorem 6.42

In the standard basis,

$$[\mathsf{T}_v]_\beta = B_v = \begin{pmatrix} \frac{1}{\sqrt{1-v^2}} & 0 & 0 & \frac{-v}{\sqrt{1-v^2}} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{-v}{\sqrt{1-v^2}} & 0 & 0 & \frac{1}{\sqrt{1-v^2}} \end{pmatrix}.$$

Proof. Columns $2, 3$ are e_2, e_3 (Theorem 6.39(a)); column 4 is $\mathsf{T}_v(e_4)$ above. Column 1 is $\mathsf{T}_v(e_1) = \frac{1}{v}(\mathsf{T}_v(v,0,0,1)^t - \mathsf{T}_v(e_4)) = \frac{1}{v}(v/\sqrt{1-v^2}, 0, 0, \sqrt{1-v^2} - 1/\sqrt{1-v^2})^t = (\frac{1}{\sqrt{1-v^2}}, 0, 0, \frac{-v}{\sqrt{1-v^2}})^t$. ■

Time contraction. A ship moving with velocity v from the solar system has S-coordinates $(vt, 0, 0, t)^t$ and S'-coordinates $(0, 0, 0, t')^t$. From $B_v(vt, 0, 0, t)^t = (0, 0, 0, t')^t$, $t' = \frac{-v^2 t + t}{\sqrt{1-v^2}} = t\sqrt{1-v^2}$: time t on Earth corresponds to $t\sqrt{1-v^2}$ on the ship. In general units, $t' = t\sqrt{1 - v^2/c^2}$.

6.10 Conditioning and the Rayleigh quotient *

Motivation. Small errors in the data b of $Ax = b$ may produce large relative errors in x. The amplification is controlled by the condition number $\|A\|\|A^{-1}\|$, which the Rayleigh quotient computes from the extreme eigenvalues of A^*A.

Example 1

$x_1 + x_2 = 5$, $x_1 - x_2 = 1 + h$ has solution $(3 + \frac{h}{2}, 2 - \frac{h}{2})$. The relative changes are $\frac{\|\delta b\|}{\|b\|} = \frac{|h|}{\sqrt{26}} = \frac{\|\delta x\|}{\|x\|}$: the system is **well-conditioned**.

Example 2

$x_1 + x_2 = 3$, $x_1 + 1.00001x_2 = 3.00001 + h$ has solution $(2 - 10^5 h,\ 1 + 10^5 h)$, so $\frac{\|\delta x\|}{\|x\|} = 10^5\sqrt{\frac{2}{5}}\,|h| \geq 10^4|h|$ while $\frac{\|\delta b\|}{\|b\|} \approx \frac{|h|}{3\sqrt{2}}$: **ill-conditioned** (the two lines are nearly parallel).

Definition (Matrix norm; Rayleigh quotient)

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, $\|A\| = \max_{x\neq 0} \frac{\|Ax\|}{\|x\|}$. For self-adjoint B, the **Rayleigh quotient** is $R(x) = \frac{\langle Bx, x\rangle}{\|x\|^2}$ $(x \neq 0)$.

Theorem 6.43

For self-adjoint B, $\max_{x\neq 0} R(x)$ is the largest and $\min_{x\neq 0} R(x)$ the smallest eigenvalue of B.

Proof. Take an orthonormal eigenbasis $\{v_i\}$, $Bv_i = \lambda_i v_i$, $\lambda_1 \geq \cdots \geq \lambda_n$ (real). For $x = \sum a_i v_i$, $R(x) = \frac{\sum \lambda_i |a_i|^2}{\sum |a_i|^2} \leq \lambda_1$, with $R(v_1) = \lambda_1$. The minimum is analogous. ■

Corollary 1

For any square A, $\|A\| = \sqrt{\lambda}$, where λ is the largest eigenvalue of A^*A.

Proof. $\frac{\|Ax\|^2}{\|x\|^2} = \frac{\langle A^*Ax, x\rangle}{\|x\|^2}$ is the Rayleigh quotient of the self-adjoint A^*A; apply Theorem 6.43. ■

Lemma

For square A, λ is an eigenvalue of A^*A iff it is an eigenvalue of AA^*.

Proof. If $\lambda \neq 0$ and $A^*Ax = \lambda x$, $x \neq 0$, then $Ax \neq 0$ and $AA^*(Ax) = \lambda Ax$; symmetrically. For $\lambda = 0$: A^*A is singular iff A is (Lemma 2 of Section 6.3), iff A^* is, iff AA^* is. ■

Corollary 2

If A is invertible, $\|A^{-1}\| = \frac{1}{\sqrt{\lambda}}$, where λ is the smallest eigenvalue of A^*A.

Proof. By Corollary 1, $\|A^{-1}\|^2$ is the largest eigenvalue of $(A^{-1})^*A^{-1} = (AA^*)^{-1}$, i.e. the reciprocal of the smallest eigenvalue of AA^*, which equals that of A^*A by the Lemma. ■

Example 3

For $A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$, $B = A^*A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ has eigenvalues $3, 3, 0$, so $\|A\| = \sqrt{3}$; indeed $R(x) = \frac{2(a^2+b^2+c^2-ab+ac+bc)}{a^2+b^2+c^2} \leq 3$.

The **condition number** of invertible A is $\text{cond}(A) = \|A\| \cdot \|A^{-1}\|$. From $\|Ax\| \leq \|A\|\|x\|$: $\|b\| \leq \|A\|\|x\|$, $\|\delta x\| \leq \|A^{-1}\|\|\delta b\|$, $\|\delta b\| \leq \|A\|\|\delta x\|$, $\|x\| \leq \|A^{-1}\|\|b\|$.

Theorem 6.44

For $Ax = b$ with A invertible and $b \neq 0$:
(a) for any norm, $\frac{1}{\text{cond}(A)} \frac{\|\delta b\|}{\|b\|} \leq \frac{\|\delta x\|}{\|x\|} \leq \text{cond}(A) \frac{\|\delta b\|}{\|b\|}$;
(b) for the Euclidean norm, $\text{cond}(A) = \sqrt{\lambda_1/\lambda_n}$, the largest and smallest eigenvalues of A^*A.

Proof.

(a) Multiply the inequalities above: $\frac{\|\delta x\|}{\|x\|} \leq \frac{\|A^{-1}\|\|\delta b\|}{\|b\|/\|A\|}$ and $\frac{\|\delta x\|}{\|x\|} \geq \frac{\|\delta b\|/\|A\|}{\|A^{-1}\|\|b\|}$.

(b) Corollaries 1 and 2. ■

Thus $\text{cond}(A) \geq 1$, with equality iff A is a scalar multiple of a unitary (orthogonal) matrix; a large condition number signals potential loss of accuracy.

6.11 The geometry of orthogonal operators *

Motivation. Every orthogonal operator on a real inner product space splits into rotations of mutually orthogonal planes and at most one reflection. In $\mathbb{R}^3$, an orthogonal operator is a rotation about an axis, possibly followed by a reflection.

Definition (Rotation; reflection)

An operator T on finite-dimensional real V is a **rotation** if $\mathsf{T} = \mathsf{I}$ or there are a 2-dimensional subspace W, an orthonormal basis $\{x_1, x_2\}$ of W and θ with $\mathsf{T}(x_1) = (\cos\theta)x_1 + (\sin\theta)x_2$, $\mathsf{T}(x_2) = (-\sin\theta)x_1 + (\cos\theta)x_2$, and $\mathsf{T}(y) = y$ for $y \in W^\perp$ (a rotation of W about $W^\perp$). T is a **reflection** if for some 1-dimensional W, $\mathsf{T}(x) = -x$ on W and $\mathsf{T}(y) = y$ on $W^\perp$ (a reflection

about $W^\perp$). Both are orthogonal.

Example 1

On a 1-dimensional real space, an orthogonal operator is $\mathsf{T} = \pm\mathsf{I}$: $\mathsf{T}(x) = \lambda x$ with $|\lambda| = 1$. So it is the identity (a rotation) or a reflection.

Example 2

(a) $\mathsf{T}(a, b) = (-a, b)$ is the reflection of $\mathbb{R}^2$ about the y-axis ($W = \operatorname{span}\{e_1\}$).
(b) $\mathsf{T}(a, b, c) = (a, b, -c)$ is the reflection of $\mathbb{R}^3$ about the xy-plane.

Theorem 6.45

An orthogonal operator T on a 2-dimensional real inner product space is a rotation or a reflection; it is a rotation iff $\det \mathsf{T} = 1$ and a reflection iff $\det \mathsf{T} = -1$.

Proof. Choose an orthonormal basis and argue as in Theorem 6.23: the matrix is $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ (a rotation, $\det = 1$) or $\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ (a symmetric orthogonal matrix with eigenvalues $1, -1$ and orthogonal eigenvectors: a reflection, $\det = -1$). ■

Corollary

On a 2-dimensional real inner product space, a composite of a reflection and a rotation (in either order) is a reflection.

Proof. It is orthogonal with determinant $(-1) \cdot 1 = -1$; apply Theorem 6.45. ■

Lemma

Every linear operator T on a nonzero finite-dimensional real vector space has an invariant subspace W with $1 \le \dim W \le 2$.

Proof [Supplement]. Factor the characteristic polynomial f over $\mathbb{R}$ into linear and irreducible quadratic factors $g_1 \cdots g_k$. By Cayley–Hamilton $f(\mathsf{T}) = g_1(\mathsf{T}) \cdots g_k(\mathsf{T}) = \mathsf{T}_0$, so some $g_j(\mathsf{T})$ is not one-to-one; take $x \neq 0$ with $g_j(\mathsf{T})x = 0$. If $g_j = t - c$, $W = \operatorname{span}\{x\}$; if $g_j = t^2 + bt + c$, then $\mathsf{T}^2 x = -b\mathsf{T}x - cx$, so $W = \operatorname{span}\{x, \mathsf{T}x\}$ is invariant. ■

Theorem 6.46

For an orthogonal operator T on a nonzero finite-dimensional real inner product space V, there are pairwise orthogonal T-invariant subspaces $W_1, \ldots, W_m$ with (a) $1 \le \dim W_i \le 2$ and (b) $V = W_1 \oplus \cdots \oplus W_m$.

Proof. Induction on $\dim V$. By the Lemma there is an invariant W_1 of dimension 1 or 2. T is injective, so $\mathsf{T}(W_1) = W_1$, and then $W_1^\perp$ is invariant: for $y \in W_1^\perp$, $x \in W_1$, $x = \mathsf{T}x'$ with $x' \in W_1$ and $\langle \mathsf{T}y, x\rangle = \langle \mathsf{T}y, \mathsf{T}x'\rangle = \langle y, x'\rangle = 0$. $\mathsf{T}_{W_1^\perp}$ is orthogonal, so induction decomposes $W_1^\perp$, and $V = W_1 \oplus W_1^\perp$. ■

Theorem 6.47

In Theorem 6.46:
(a) the number of W_i on which T_{W_i} is a reflection is even or odd according as $\det \mathsf{T} = 1$ or -1;
(b) the decomposition can be chosen so that this number is 0 or 1 according as $\det \mathsf{T} = 1$ or -1, and so that every 1-dimensional W_i on which T is a rotation (i.e. the identity) is orthogonal to all the others.

Proof.

(a) $\det \mathsf{T} = \prod \det \mathsf{T}_{W_i}$ (Theorem 5.25: block-diagonal matrix), and $\det \mathsf{T}_{W_i} = -1$ exactly for reflections (Example 1, Theorem 6.45).

(b) Split each 2-dimensional W_i on which T is a reflection into the two orthogonal lines E_1, E_{-1} of T_{W_i}. Now all reflection pieces are lines with $\mathsf{T} = -\mathsf{I}$; pair them up two at a time, each pair spanning a plane on which $\mathsf{T} = -\mathsf{I}$, a rotation by π. At most one line with $\mathsf{T} = -\mathsf{I}$ remains, and its presence is decided by (a). ■

Corollary

Every orthogonal operator T on a finite-dimensional real inner product space is a product $\mathsf{T} = \mathsf{T}_1 \cdots \mathsf{T}_m$ of commuting orthogonal operators T_i, each a rotation or a reflection, at most one of them a reflection.

Proof. With $V = W_1 \oplus \cdots \oplus W_m$ as in Theorem 6.47(b), let $\mathsf{T}_i = \mathsf{T}$ on W_i and $\mathsf{T}_i = \mathsf{I}$ on $W_i^{\perp} = \bigoplus_{j \neq i} W_j$. Each T_i is a rotation or a reflection, they commute (they act on different summands), and $\mathsf{T}_1 \cdots \mathsf{T}_m$ agrees with T on every W_i. ■

Example 3

For orthogonal T on a 3-dimensional space, Theorem 6.47(b) gives $m = 2$ (a line W_1 and a plane W_2, with T_{W_2} a rotation) or $m = 3$ (three lines). In the first case T is a rotation about the axis W_1, followed by a reflection in $W_1^{\perp}$ if $\mathsf{T}_{W_1} = -\mathsf{I}$. In the second, at most one line carries $\mathsf{T} = -\mathsf{I}$, so T is the identity or a reflection. Hence: an orthogonal operator on $\mathbb{R}^3$ is a rotation about an axis or such a rotation composed with a reflection, according as $\det \mathsf{T} = 1$ or -1.